MAGNETOM

Replacement of Parts
System

Gradient

Symphony, A Tim System

Document Version

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1.1 General

The following describes the replacement procedures of the "Field replaceable Units" (FRU's) in the gradient system K2217 (KAS) and provides references for the checks and adjustments required.

Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the "sub-component" (e.g. front cover) and the protective earth wire connected to the "main component" (e.g. GPA frame).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

Tab. 1 FRU's GPA K2217

Assembly	Time	Adjustments required	Final Checks	Repair Instructions	Required Tools
GSSU/D11(2) DAC Board (73 69 973 replaces 57 25 507 and 59 41 534)	85 min.	▪ Check jumper/switch position, check offset. ▪ TuneUp/ Phantom Shim, Gradient Sensitivity, Gradient Delay 6 min.	▪ QA/ All selected procedures 29 min.		
GSSU/D12 Regulator - Board (57 26 349)	160 min.	▪ Check jumper/ switch positions, check SENS offset. ▪ TuneUp/ Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Sensitivity, Gradient Delay 50 min.	▪ QA/ All selected procedures 68 min.	(→ GSSU Components (E5) K2217 / Page 15)	▪ Standard tool set ▪ Ground test meter ▪ DVM Fluke 187 (material number 99 94 831 or equivalent) ▪ ESD equipment (097 02 606)
GSSU/D14 Modulator - Board (57 26 364)	160 min.	▪ Check jumper setting. ▪ TuneUp/ Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Sensitivity, Gradient Delay 50 min.	▪ QA/ All selected procedures ▪ MR image 68 min.		
GSSU/D16 CAN-Service Board (57 25 515)	45 min.	▪ Check jumper setting.	▪ QA/ Regulator Check, Coil Check		
GSSU/D17 I/O - Board (57 26 430)	45 min.	▪ Check jumper setting/switch positions.	▪ QA/ Regulator Check ▪ MR image 12 min.		

Assembly	Time	Adjustments required	Final Checks	Repair Instructions	Required Tools
GSSU/D139 CAN & Service Board (83 65 981)	45 min.	■ Check jumpers	■ MR image 5 min.	(→ GSSU Components (E5) K2217 / Page 15)	■ Standard tool set ■ Ground test meter ■ ESD equip-ment (097 02 606)
GSSU Power Supply (E4A) (30 59 482)	75 min.	■ Check / Adjustment of output voltages.	■ MR image 5 min.	(→ GSSU Power Supply (E4A) / Page 24)	■ Standard tool set ■ Ground test meter ■ DVM Fluke 187 (material number 99 94 831 or equivalent)

Assembly	Time	Adjustments required	Final Checks	Repair Instructions	Required Tools
Current Sen-sors (107 43 890)	295 min.	▪ Check SENS offset adjustment on D12 ▪ TuneUp/ Regulator Adjust, Table Adjustment, Phantom Shim, CTC, ECC, Gradient Sensitivity, Gradient Delay 50 min.	▪ QA/ All selected procedures, Image Orientation Check, Regulator Check, Shim Check, CTC Check, ECC Check, Grad Sens Check, Gra-dient Delay Check, Calc Arti-facts, Spike Check, Stability Check, FatSat Check 68 min.	(→ Current Sensor As-sembly / Page 36)	▪ Standard tool set ▪ Ground test meter ▪ SW19 open-ended wrench with extension and ratchet ▪ Torque wrench 20-50 Nm (material number 83 95 803) ▪ Accessory for torque wrench (ma-terial num-ber 83 96 116) ▪ 10 mm, 4-mm Allen wrenches ▪ Torx 25 bit ▪ Special tools for gradient connection (material number 105 63 214)

Assembly	Time	Adjustments required	Final Checks	Repair Instructions	Required Tools
Power Stage (106 63 308)	155 min.	■ TuneUp/ Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Sensitivity, Gradient Delay 50 min.	■ QA/ All selected procedures, Image Orientation Check 68 min.	(➔ Power Stage / Page 50)	■ Standard tool set ■ Ground test meter ■ SW 19 wrench Note: D32 + fuses potentially defect as well (see description)
Choke (57 25 887)	190 min.	■ TuneUp/ Regulator Adjust, Phantom Shim, CTC, ECC, Gradient Sensitivity, Gradient Delay 50 min.	■ QA/ All selected procedures, Image Orientation Check 68 min.	(➔ Chokes of GPA K2217 / Page 77)	■ Standard tool set ■ Ground test meter Note: Always replace the 2 chokes of one gradient axis circuit together. Replace the corresponding Power Stage as well.

Assembly	Time	Adjustments required	Final Checks	Repair Instructions	Required Tools
Blower (30 63 476)	115 min.	■ No action necessary	■ Check blower function ■ QA/ Image Orientation Check 2 min.	(→ Blowers of GPA K2217 / Page 82)	■ Standard tool set ■ Ground test meter Note: All three Power Stages must be re-moved.
Rectifier D32 (83 64 197)	50 min.	■ No action necessary	■ QA/ Regulator Check ■ MR image 10 min.	(→ Rectifier D32 K2217 / Page 60)	■ Standard tool set
Soft Start D31 (57 26 414)	60 min.	■ No action necessary	■ MR image 5 min.	(→ Soft Start D31 K2217 / Page 65)	■ Standard tool set
Main Contact. K2 (30 90 560)	80 min.	■ No action necessary	■ MR image 5 min.	(→ Main Contactor K2 GPA K2217 / Page 69)	■ Standard tool set ■ Torx 25 driv-er bit
Line Filter (57 25 671)	70 min.	■ No action necessary	■ MR image 5 min.	(→ Line Filter of GPA K2217 / Page 30)	■ Standard tool set ■ Ground test meter
Water Distrib-utor (73 70 146)	195 min.	■ No action necessary	■ Check tightness of connec-tions and search for leaks ■ MR image 5 min.	(→ Water Distribu-tor / Page 87)	■ Standard tool set

Assembly	Time	Adjustments required	Final Checks	Repair Instructions	Required Tools
Cabling	10 min.	■ No action necessary	■ QA/ Image Orientation Check 2 min.	n.a.	■ Standard tool set
Water Distributor Gradient Coil (59 38 688)	150 min.	■ No action necessary	■ Check water distributor and water connections for leaks. ■ MR image 5 min.	(→ GC Water Distributor / Page 90)	■ Non-magnetic standard tool kit (77 57 706) ■ Non-magnetic scalpel
Gradient cable kit Sonata (11 41 923)	180 min.	■ All connections must be tightened with the specified torque, using the listed non-magnetic torque wrenches	■ Double-check and ensure that all connections are tight ■ QA/ Spike Check, Image Orientation Check	(→ Replacing the Gradient Cables / Page 96).	■ Non-magnetic standard tool kit (77 57 706) ■ Non-magnetic torque wrench ranging from 5 - 20 Nm (83 95 829) ■ Non-magnetic torque wrench ranging from 20 - 50 Nm (83 95 803) ■ Non-magnetic socket assortment (83 96 116)

Tab. 2 Gradient Supervision

Gradient Supervision VLM					
FRU / Material number	Total Time	Instructions	Adjustments	Function Test	Tools
Aspirating_Smoke_Detector_VLM-000-S (106 83 917) or Smoke_Detector_Configur._VLM-001-SIE (with higher alarm threshold for countries with strong environmental pollution) (109 15 504)	90 min.	Please refer to: **CB-DOC/Installation/Startup/System Start-up/Startup of the Gradient Supervision** (→ Start-Up Procedure of the VLM Device / M6-015.815.01)			▪ Standard tool set

WARNING

Troubleshooting and repairs involving current sensors, gradient filters, gradient coil, gradient cables might lead to problems with the system's image orientation.

This could result in a misdiagnosis.

» Perform the "Image Orientation Check" after all repair procedures when you have to disconnect or open the gradient coil power circuit, e.g. replacement of: Current Sensors, Gradient Filters, Gradient Coil, Gradient Cables.

CAUTION

High Voltage across capacitor/gradient cable connections.

Risk of electrical shock!

» After turning off the GPA cabinet, the voltage decreases to a safe level after 20 minutes.

CAUTION

Hot components

Particularly after a heavy load, the output filter chokes, the transformer and rectifier circuits can get very hot. Touching these parts can cause severe skin burn.

» Do not touch these parts until at least 30 min. after you have shut down the GPA.

Adhere to the general safety guidelines in the safety chapter.

Never unplug the quick connectors of a water-cooled component when the water pump in the RCA is in operation (because of high impact pressure). Switch off the water pump via the circuit breaker F1 in the RCA cabinet.

Before returning the replaced power stage, drain the water from the pipes by opening both ends.

Before returning the replaced rectifier board D32, drain the water from the pipes by opening both ends.

3.1 GSSU Components (E5) K2217

3.1.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

3.1.2 Tools and Time required

3.1.2.1 Time

Tab. 3 Completion Time

PCB	Time	Persons
DAC Board D11(2)	85 min.	1
Regulator D12	160 min.	1
Modulator D14	160 min.	1
CAN-Service D16	45 min.	1
CAN & Service Board D139	45 min.	1
I/O - Board D17	45 min.	1

3.1.2.2 Tools

- Set of standard tools
- Ground test meter
- DVM Fluke 187 (material number 99 94 831) or equivalent
- ESD equipment (097 02 606)

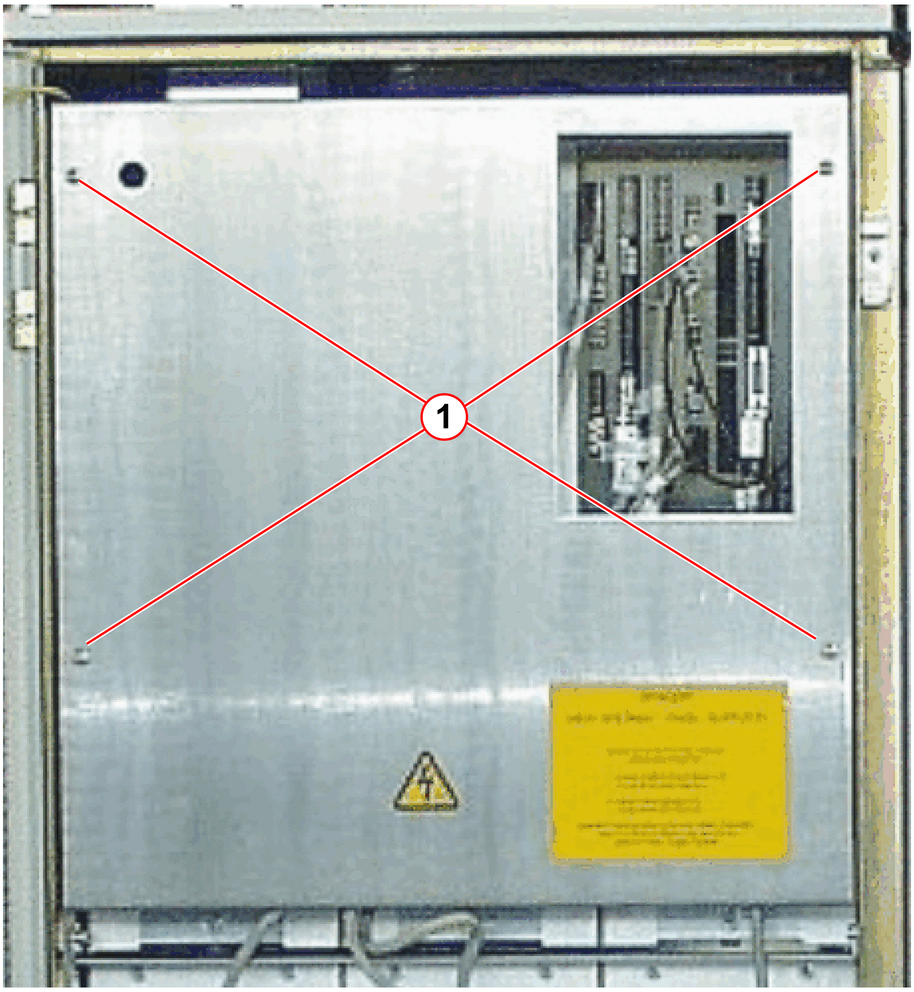
  Adhere to ESD regulations and use the appropriate tool set: Anti-static wristband and grounded pad

3.1.3 Preliminary

- Switch off circuit breaker F6 (in the ACC) and F110 on top of the ACC.
- Wait at least for 20 minutes to ensure that the final stage capacitors are fully discharged.

- Remove GSSU front cover (4 screws).

Fig. 1: GSSU - front cover



(1) Screws

Position of the PCBs:

D11(2) - D12 - D14 - D16 - D139 - D17

Fig. 2: Gradient Small Signal Unit (GSSU)

3.1.4 Disassembly

- Remove the connectors and FOCs of the defective board.
- Remove the board from the frame.

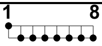
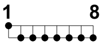
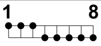
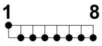
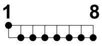
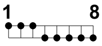
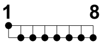
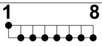
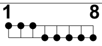
> To pull GSSU boards easily, you may insert a cable tie or a wire loop through the holes of the board front panel.

3.1.5 Assembly

3.1.5.1 Jumper Settings

- Check / set jumpers and DIP switch positions before installing new boards.

Fig. 3: Jumper Settings

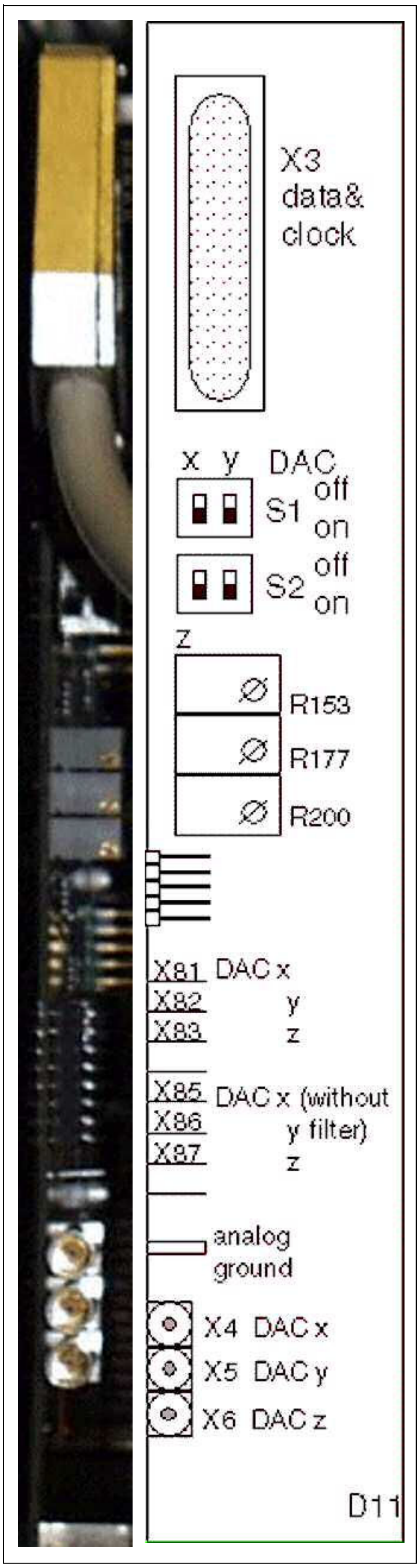
E5 K2217 : GSSU		
PCB	**Jumper- & switch-setting**	
D11(2)	**X16**	**2 - 3**
	S1	**both down**
	S2	**both down**
	S3	**3**
	S4	**6**
D12	**S1**	**ON 1 8** 
	S2	**ON 1 8** 
	S3	**ON 1 8** 
	S4	**ON 1 8** 
	S5	**ON 1 8** 
	S6	**ON 1 8** 
	S7	**ON 1 8** 
	S8	**ON 1 8** 
	S9	**ON 1 8** 
	X16	**1 - 2**
	X17	**1 - 2**
	X18	**1 - 2**
	X19	**1 - 2**
	X20	**1 - 2**
	X21	**1 - 2**
	X22	**1 - 2**
	X23	**1 - 2**
	X24	**1 - 2**
	X25	**18 - 19**
	X25	**21 - 22**
	X25	**24 - 25**
D14	**X9**	**1 - 2**
	X10	**1 - 2**
	X11	**6 - 7**
	X11	**18 - 19**
	X12	**1 - 2**
	X13	**1 - 2**
D16	**X3**	**1 - 2**
	X601	**1 - 2**
D139	**X3**	**1 - 2**
D17	**X9**	**11 - 12**
	X9	**19 - 20**
	S1	**both down**
	S2	**both down**

- Connect all relevant connectors (FOCs, signal/power lines).

3.1.5.2 DAC D11(2) Offset Adjustment

Prerequisite is a performed regulator adjustment (TuneUp).

Fig. 4: Front of DAC board D11



Descriptions:

X3:

20 bit parallel gradient data (18 bit used) and 4MHz clock signal from AMC

Gradient data: D11/X3 - AMC/D5/X11

S1/S2:

The DAC output can be switched on/off for every axis; S2/2 is not used.

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V) 10 V corresponds to the maximum output current.

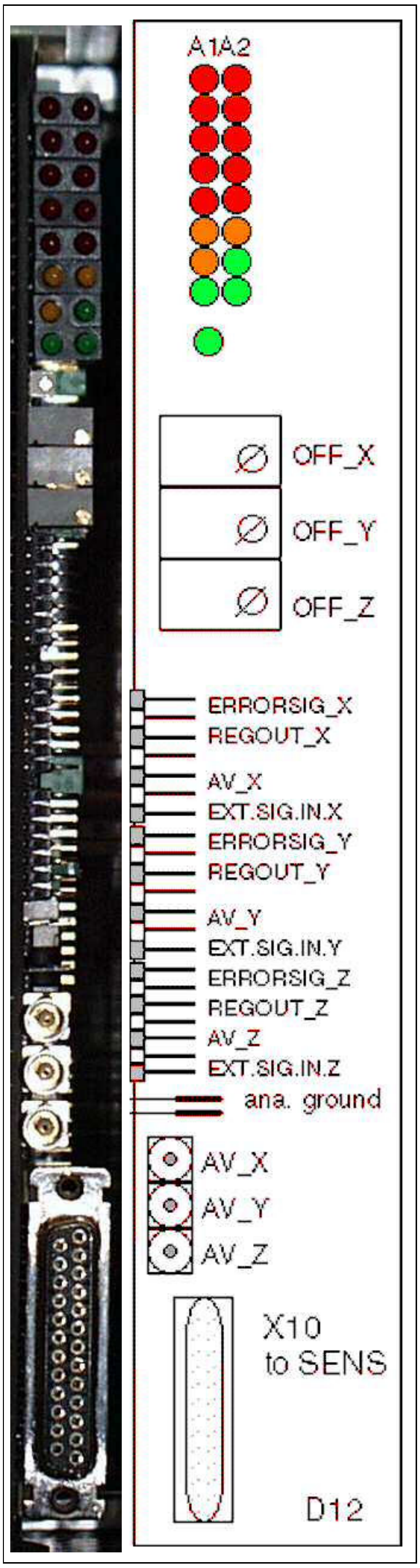
- Gradient cabinet has been on for at least 10 minutes.
- Set switches S1/2 to up position ("DAC output zero").

- The corresponding modulator must be on: S1/2 on D4 in down position.
- Connect a DVM to the desired actual value output X11,12,13 on D12. Use the QLA-BNC service cable.
- Check the offset: Offset < +/- 0.05 mV.
- Use the corresponding R153, 177, 200 on DAC-board D11(2) to adjust the offset for the X, Y and Z axes if necessary.

3.1.5.3 Regulator D12 SENS Offset Adjustment

Prerequisite is a performed regulator adjustment (TuneUp).

Fig. 5: Front of Regulator D12



Descriptions:

LEDs (A1 left / A2 right):

V130: IMAX_X / IMAX_Y V131: IMAX_Z / CAV_ERR_X

V132: CAV_ERR_Y / CAV_ERR_Z

V133: DUCY_X / DUCY_Y

V134: DUCY_Z / OHMLOSS

V135: LOOP_ERR_X / LOOP_ERR_Y

V136: LOOP_ERR_Z / REG_ON_X

V137: REG_ON_Y / REG_ON_Z

V159: current flow

Offset pots

X: R 155

Y: R 189

Z: R 223

Test points (refer to Function Description)

X11,12,13: Actual current value X,Y,Z QLA (1V=50A)

X10 SENS connection

- Gradient cabinet has been on for at least 10 minutes.

- Connect the DVM to X11,12,13 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the I/O board D17 in up position ("Modulator off").
- Check the offset: Offset < +/- 0.05 mV; use offset pots to adjust the offset if necessary.

3.1.6 Concluding Steps

- Switch on F110 and then F6.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC. Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Perform 'Reboot Scanner'.
- Check that the blower at GSSU Power supply E4A and GSSU top is functioning properly.
- Install the front cover.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.

3.1.7 Adjustments

- Perform TuneUp steps according (→ Tab. 1 Page 7).

3.1.8 Final Checks

- Perform QA steps according (→ Tab. 1 Page 7).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

3.2 GSSU Power Supply (E4A)

(Material number 30 59 482)

3.2.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

3.2.2 Tools and Time required

3.2.2.1 Time

75 min.

3.2.2.2 Tools

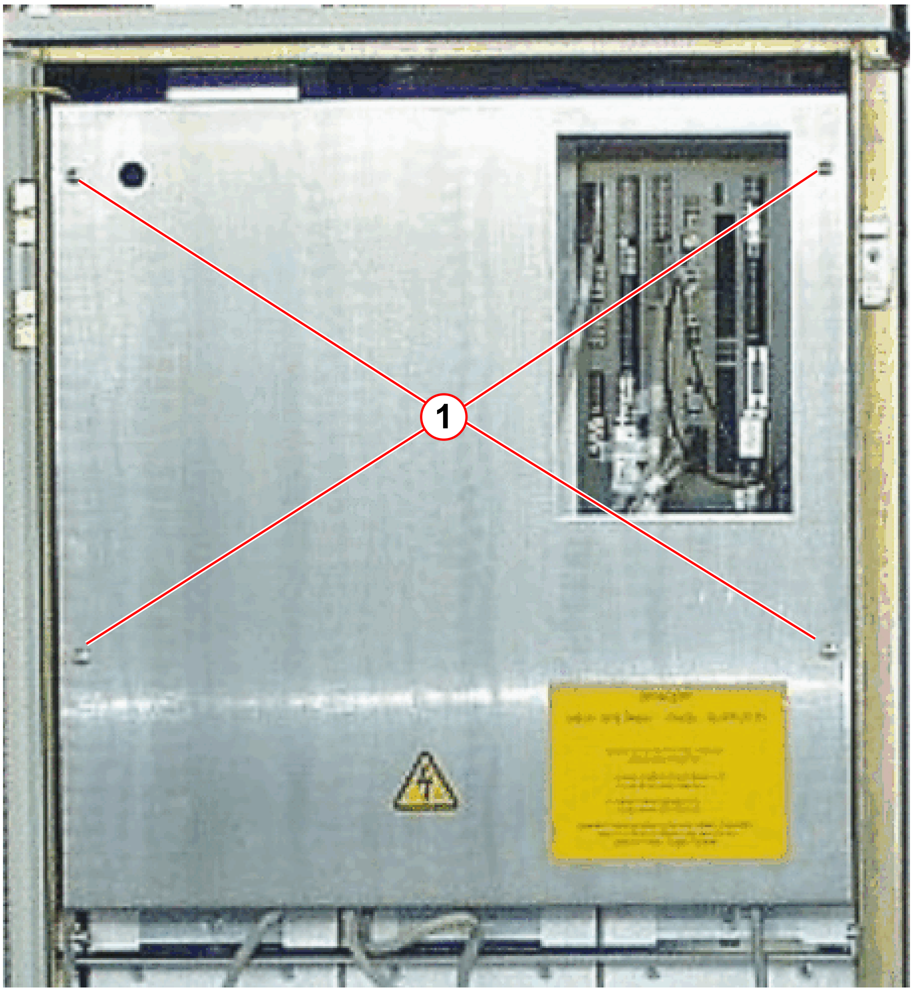
- Set of standard tools
- Ground test meter
- DVM Fluke 187 (material number 99 94 831) or equivalent

3.2.3 Preliminary

- Switch off circuit breaker F6 (in the ACC) and F110 on top of the ACC.
- Wait at least for 20 minutes to ensure that the final stage capacitors are fully discharged.

- Remove GSSU front cover (4 screws).

Fig. 6: GSSU - front cover



(1) Screws

3.2.4 Disassembly

- Disconnect all Faston connectors from the power supply unit front side.
- Push the cables away.
- Loosen, but do not remove the 4 screws (3 mm Allen key) at the bottom front and rear side.
- Slide the power supply unit to the front.
- Disconnect the Faston connectors of the AC line voltage cable at power supply unit rear-side.

3.2.5 Assembly

- Install the GSSU power supply unit in reverse order of removal.
- Connect the AC line voltage cable rear-side.

- Connect the DC cables front-side.

3.2.6 Concluding Steps

- Switch on F110 and then F6.

> **NOTICE**
>
> **Switching on the GPA circuit breakers in wrong sequence can damage GPA.**
>
> **Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC. Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!**
>
> » **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Perform 'Reboot Scanner'.

3.2.7 Adjustments

> Measure / adjust the voltages with disabled modulators (S1 left/right and S2 left at D17 set to upward position).
>
> All PCBs of the GSSU must be installed in the rack.

Fig. 7: Adjustment of supply voltage

Slot 1: +5.075 ± 0.025 VDC

Measure the voltage on D17 (➔ Fig. 8 Page 28).

Slot 2: +16,5 ± 0.100 VDC

Supply voltage for the drivers of the Power Stages and fan on top of GSSU.

Measure the voltage on (➔ Tab. 4 Page 27).

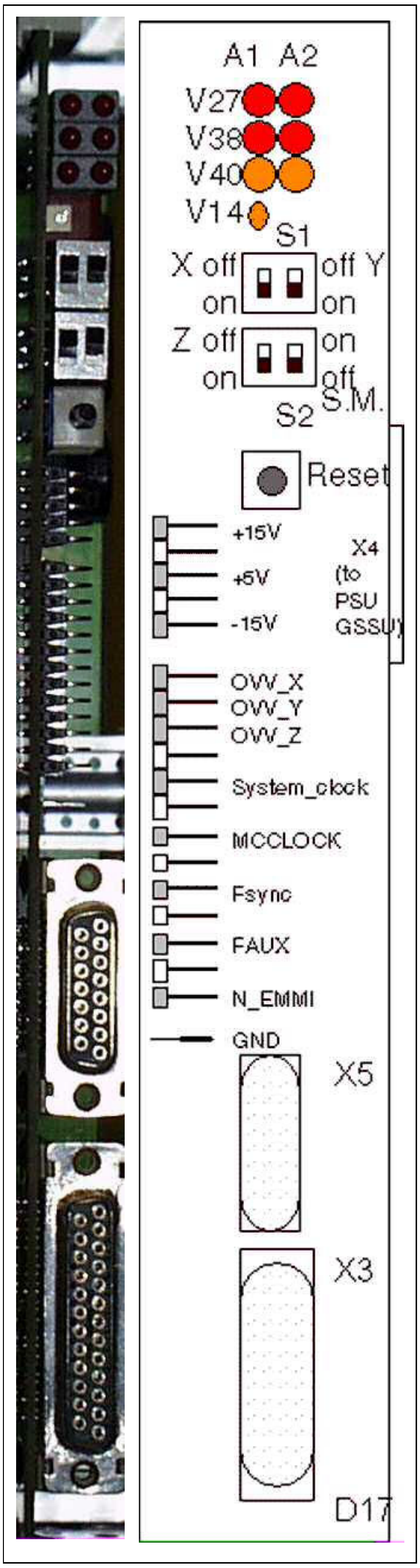
Slot 3: +15.05 ± 0.025 VDC

Measure the voltage on (➔ Fig. 8 Page 28).

Tab. 4 Voltage Terminals at GSSU Power Supply E4A

Slot	Terminal at E4A
Slot 1	A = 5V, B = GND, C = 5V (Sense), D = GND (Sense)
Slot 2	E = +Fan, F = GND Fan, G = +Driver supply, H = GND Driver supply
Slot 3	I = +15V, J = GND, K = GND, L = -15V

Fig. 8: Front of I/O board D17



3.2.8 Final Checks

- Install the front cover.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.
- Measure an MR image

  5 min.

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

3.3 Line Filter of GPA K2217

(Material number 57 25 671)

3.3.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

3.3.2 Tools and Time required

3.3.2.1 Time

70 min.

3.3.2.2 Tools

- Standard tool set
- Ground test meter

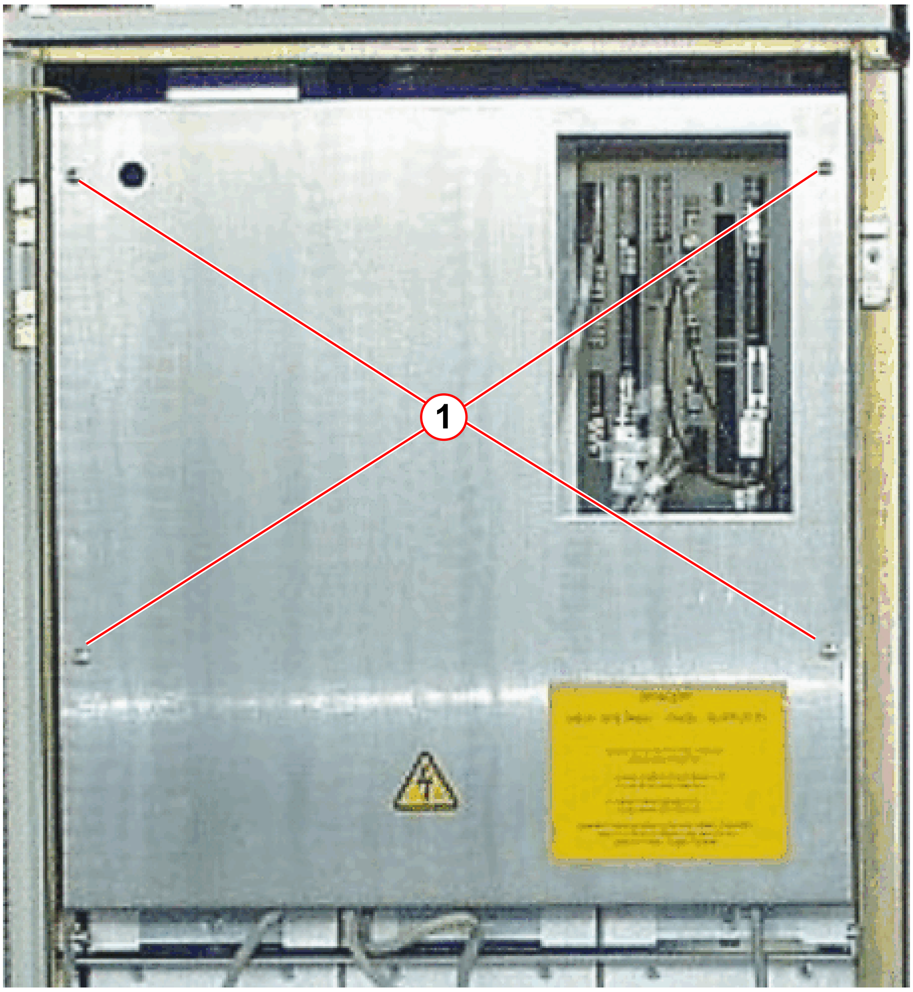
3.3.3 Preliminary

- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait at least for 20 minutes to ensure that the final stage capacitors are fully discharged.

3.3.4 Disassembly

- Remove GSSU front cover (4 screws).

Fig. 9: GSSU - front cover



(1) Screws

- Remove the line cables at the connectors on the bottom of the filter.
- Remove the cover on top of the GPA cabinet (➔ Fig. 11 Page 33).
- Remove the input line cables on top of the filter (➔ Fig. 12 Page 34).
- Loosen the four screws on top of the GPA cabinet (➔ Fig. 12 Page 34).
- Remove the nuts at the right side which clamp the filter housing to the GPA cabinet (via a current rail) (➔ Fig. 10 Page 32).
- Take out the line filter.

Fig. 10: Line filter mounted in the upper part of the GPA cabinet

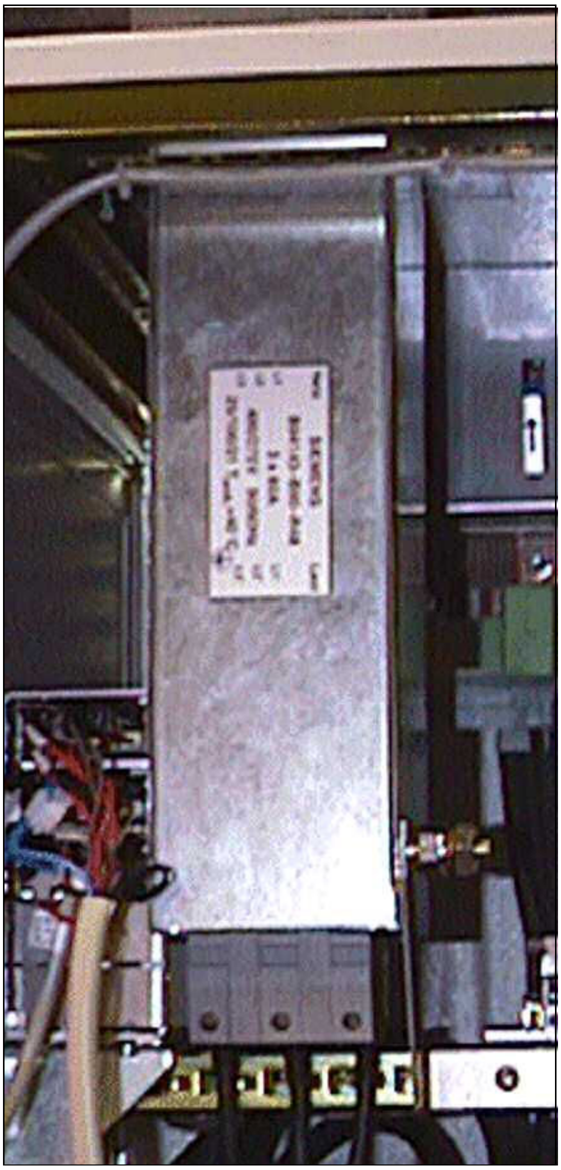


Fig. 11: Cover of line filter connections on top of GPA cabinet

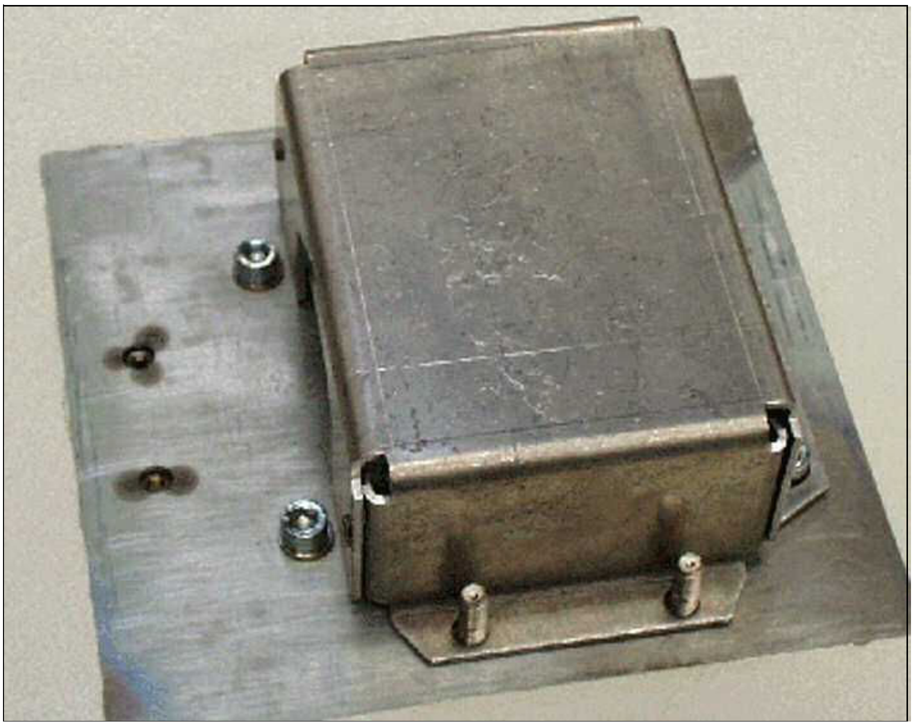


Fig. 12: Line cable connection on top of line filter

3.3.5 Assembly

- To install the line filter, proceed in reverse order.

3.3.6 Concluding Steps

- Install the GSSU front cover.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» First switch on circuit breaker F110 / mains box, then F6/ ACC.

- Switch on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

3.3.7 Final Checks

- Measure an MR image.

Time 5 min.

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

4.1 Replacing the Current Sensor Assembly K2217

(Current Converter Unit 059: 107 43 890)

4.1.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts will lead to death or serious physical injury.

» **Switch off the gradient cabinet with circuit breakers. (Both GPA power stage supply F110/ mains box and GSSU power supply F6/ ACC)**

WARNING

High voltage is still present on the final stage output terminals!

Contact with voltage conducting parts leads to death or serious physical injury.

» **After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.**

Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

Check the green LED's behind the 6 holes in the PS front side (→ Fig. 29 Page 53). If the LED's are on, the capacitors are charged! Later final stages may not have the LEDs fitted any more!

4.1.2 Estimated Completion Time and Number of Persons

Tab. 5 Completion Time

Work Steps	Time	Persons
Preparations	20 min.	1
Replacement / Installation	155 min	1
Startup/TuneUp/QA	120 min.	1
Final Steps	n.a..	1

4.1.3 Aids or Tools

- Standard tool set

- Ground test meter
- SW 19 open-ended wrench with extension and ratchet
- Torque wrench 20-50 Nm (material number 83 95 803)
- Accessory for torque wrench (material number 83 96 116)
- 10 mm, 4-mm Allen wrenches
- Torx 25 bit
- Special tools for gradient connection (material number 105 63 214)

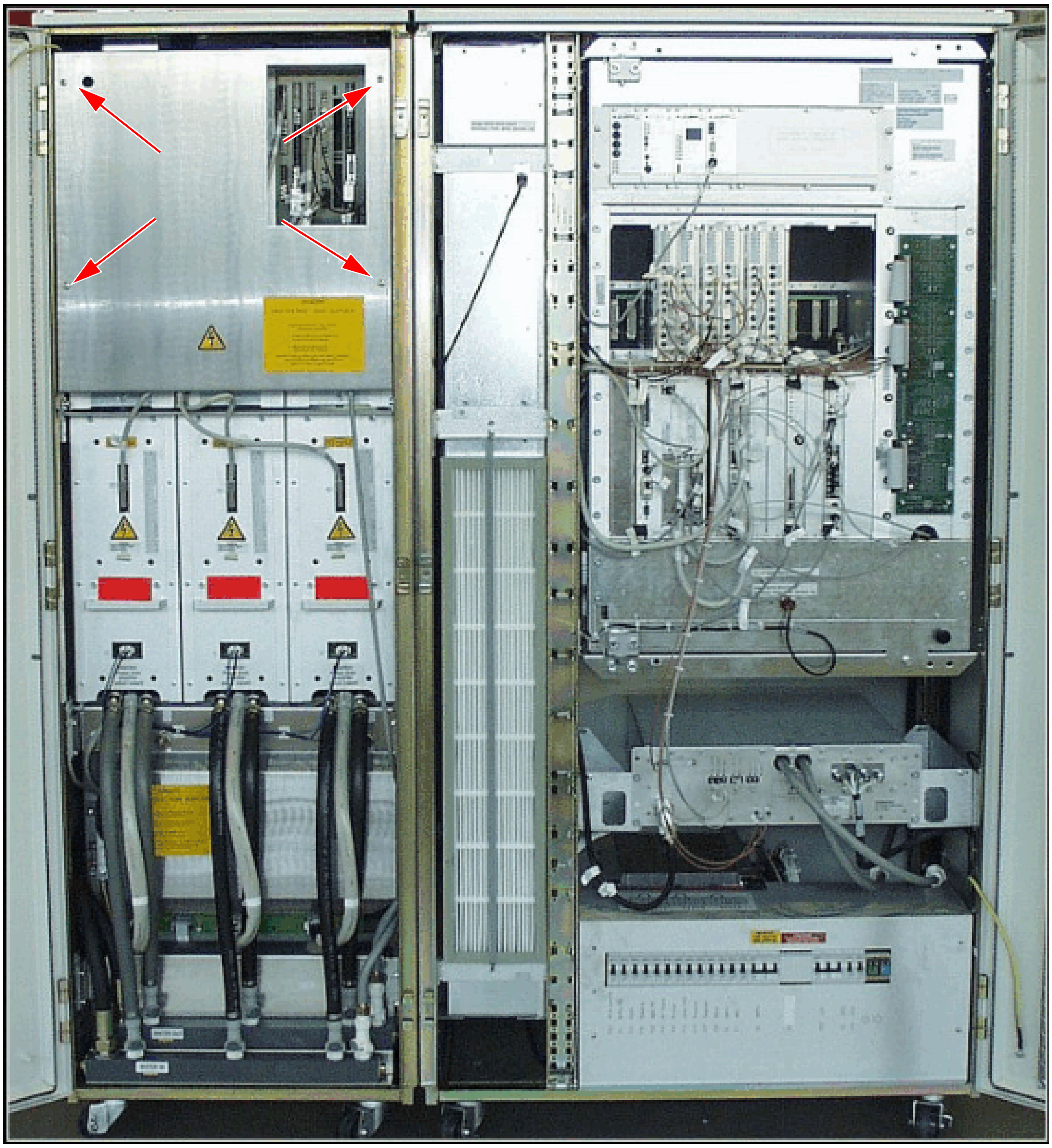
4.1.4 Preparation

- Switch off the transformer T1 by pressing S403 at the D16/GSSU.
- Switch off circuit breaker F6 (in the ACC) first, then F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.

4.1.5 Removing/installing the Current Sensor Assembly

- Remove the cover plate in front of the GSSU; 4 quick lock screws.

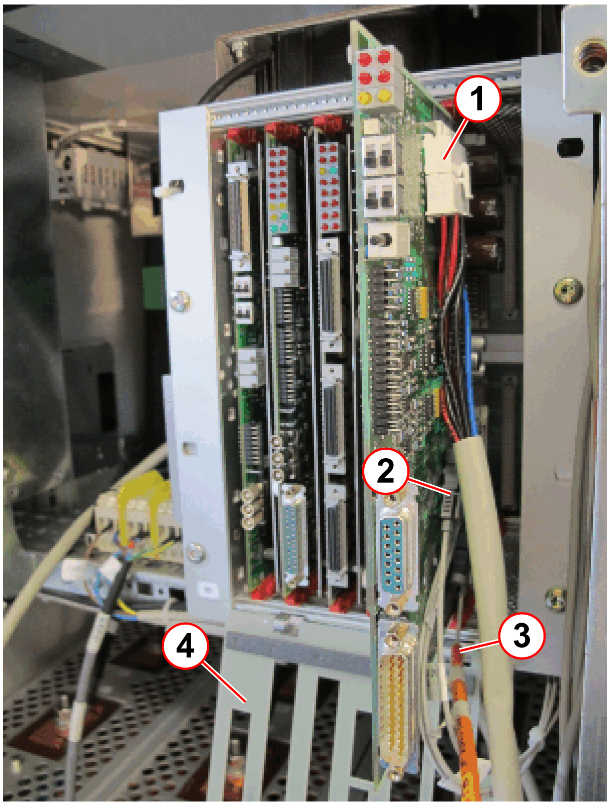
Fig. 13: GSSU cover plate



4.1.5.1 Removing the GSSU

For GSSU removal, you do not have to pull out any boards (except D17- for removing connector X4 and FOC's). Use the ESD kit.

Fig. 14: D17 in the GSSU



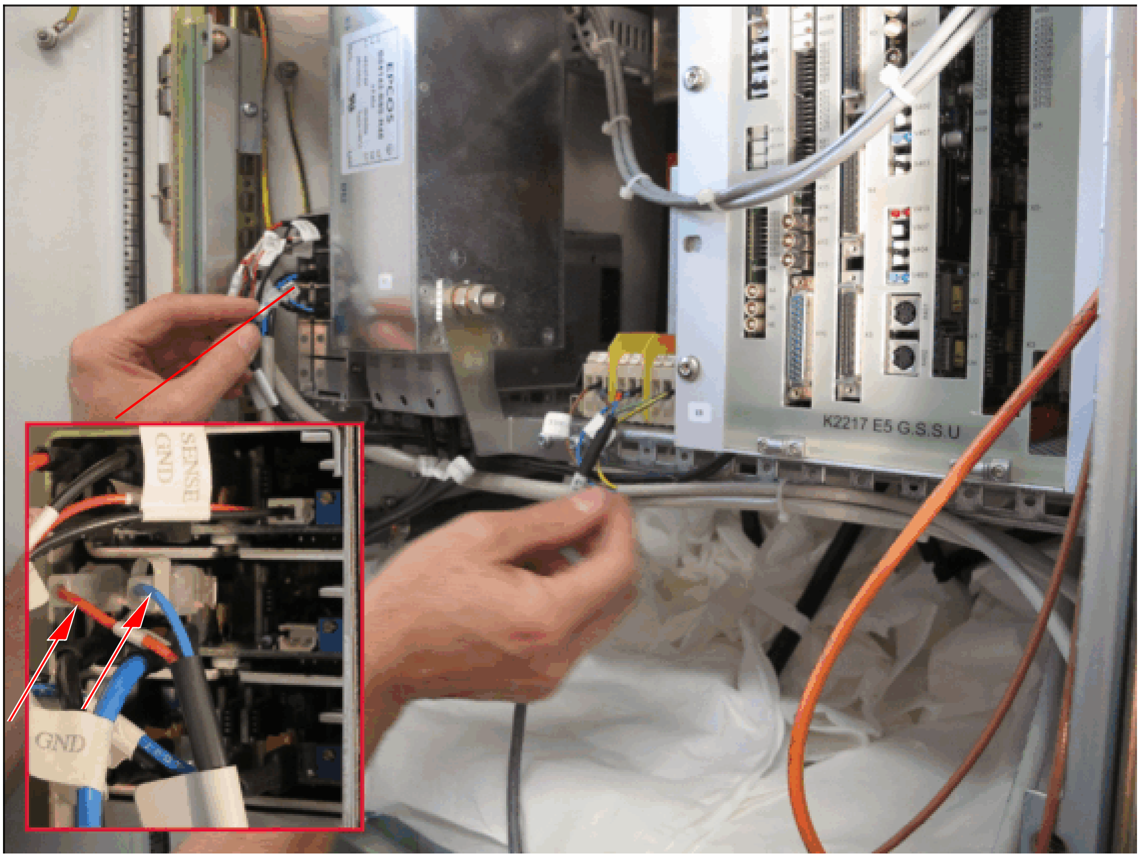
(1) X4
(2) FOC's U1, U2, U3
(3) U7
(4) Cover plate GSSU

- Remove all signal cables and FOCs from the GSSU boards
- Remove the cover plate of the GSSU (Pos. 4).
- Remove the board D17 carefully and disconnect X4 (Pos. 1) and FOC's U1,2,3 (Pos. 2) to the current sensor assembly and U7 (Pos. 3)

- The GSSU fan must be disconnected from the GSSU power supply unit
- Cut the cable ties of the fan power cable.

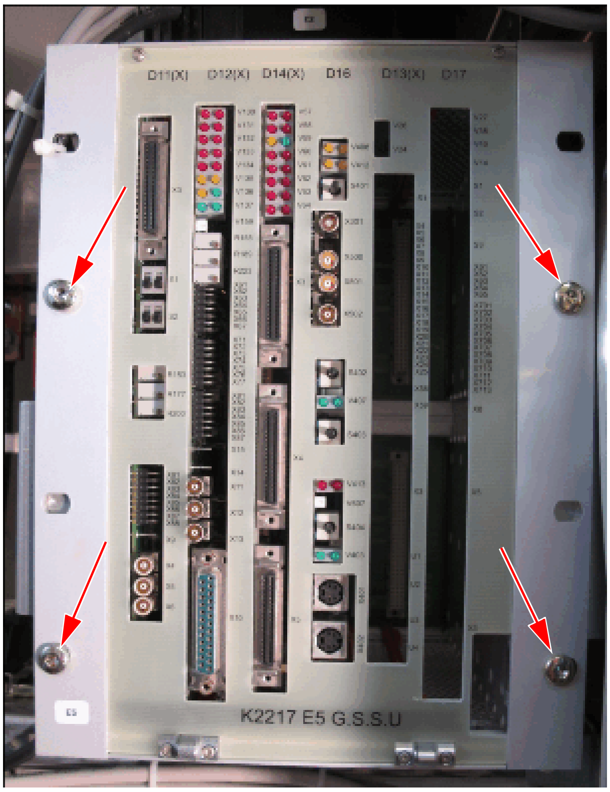
- Disconnect the fan cable from the GSSU power supply unit (➔ Fig. 15 Page 40)

Fig. 15: Fan power connection



(1) Power lines fan

Fig. 16: GSSU disconnect



(1) quick lock screws

Remove the GSSU completely to access the cable connections to the current sensor assembly..

- Loosen the quick lock screws.(Pos. 1)
- Remove the complete GSSU rack (together with the fan) and store it safely.

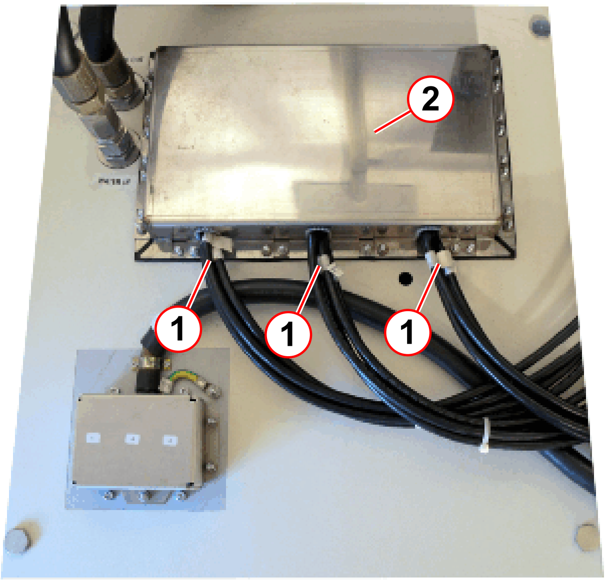
4.1.5.2 Connection at GPA roof-top

The type of gradient cable and cover depends on the location of the cabinet (either in the room or directly underneath the RF-filter plate).

For systems with the short cable set, the gradient connection screws at the GPA roof are difficult to reach. Use an extra short-head 10-mm Allen key. Special tools for this connection can be also ordered from SPC. These tools are fully magnetic and must be only used in the equipment room.

Shorter Cable Set 2.5 m - horizontal cable outlet unshielded version.

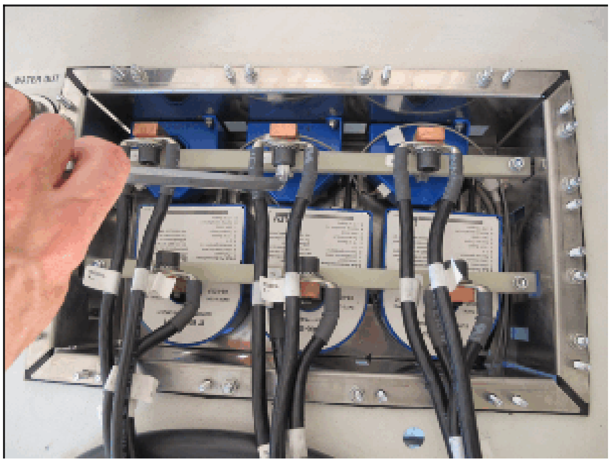
Fig. 17: gradient cable on top of the GPA cabinet (short cable set)



(1) Gradient cables
(2) cover of the gradient cable

- Remove the nuts M6 for the gradient cover to get access to the gradient connection. Use a socket wrench.

Fig. 18: Gradient connection on top of the GPA (short cable set)



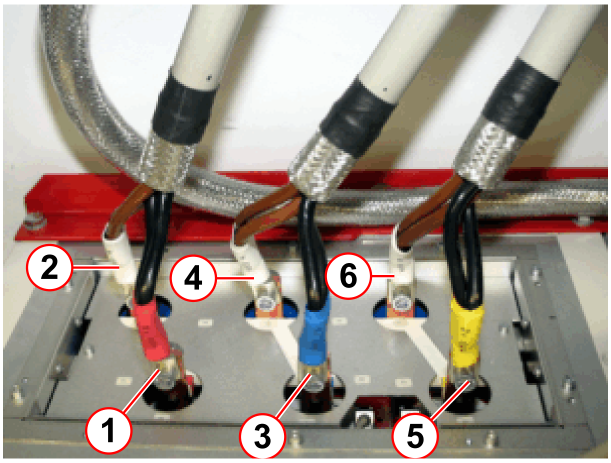
- Loosen the gradient connection screws with a short-head 10- mm Allen key (included in special tools for gradient connection, material number 10563214). .
- Remove the nuts M6 nuts of the current sensor unit.

Long cable Set- vertical cable outlet

Fig. 19: Gradient cables on top of the GPA cabinet (long cable set)

- Remove the GPA top cover (including the cable clamps).

Fig. 20: GPA gradient cable connection



(1) X+
(2) X-
(3) Y+
(4) Y-
(5) Z+
(6) Z-

- Remove the gradient cable connection screws.

4.1.5.3 Disconnecting the Gradient Cables in the GPA

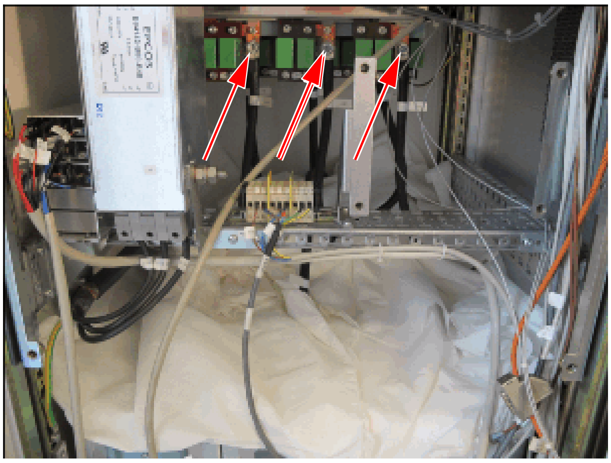
NOTICE

If a metallic part (e.g. nut, washer) drops into a power stage, remove it !

Each loose metallic particle in the power stage is highly dangerous and may cause a short circuit under power !

- » **Cover the power stages with a suitable material like a blanket, paper towel, etc.**
- » **In case a part has dropped into a power stage turn it upside down and shake it carefully until the missing part falls out.**

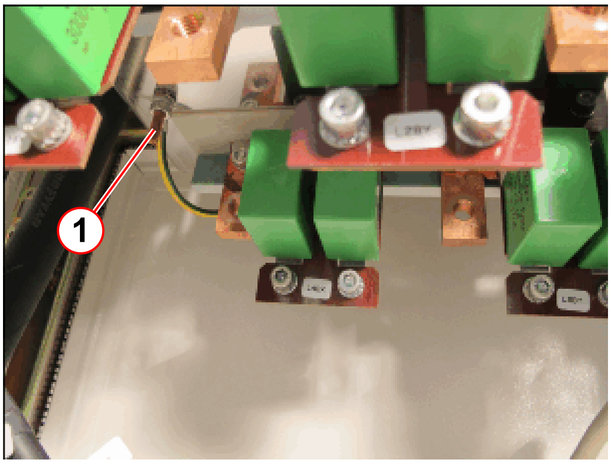
Fig. 21: Protected power stages



(1) Front gradient connection

- Loosen the screw connections (M10 x 8-mm hex socket head) of the gradient cables.

Fig. 22: Grounding of the Current sensor unit

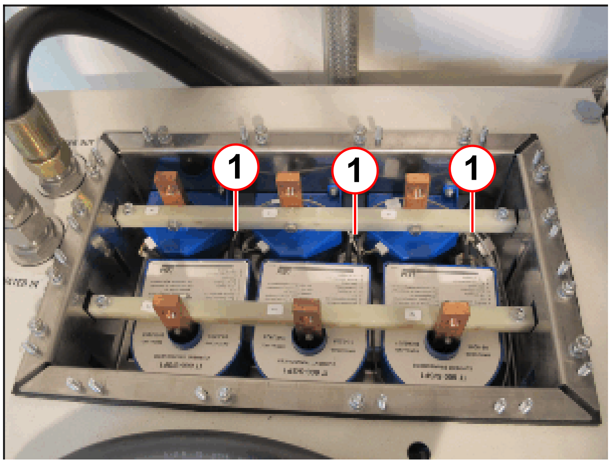


(1) Grounding bolt

- Loosen the ground wire at the current sensor unit inside the GPA cabinet. (Pos. 1)

Disconnect the 3 FOCs at the current sensor. These FOCs are not used for LEM current sensor material number 10743890.

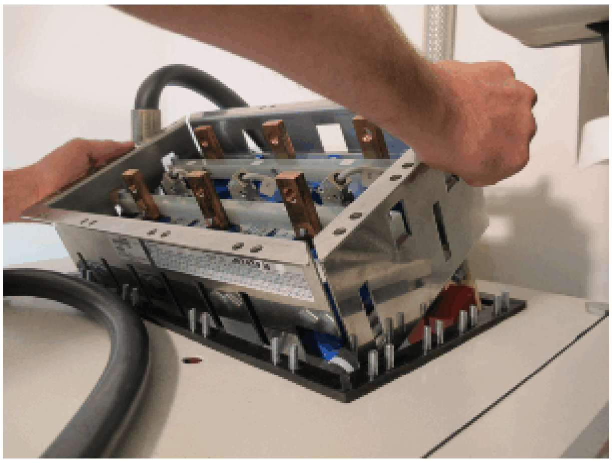
Fig. 23: LEM with FOC's



(1) FOC

- Remove the 3 FOC's (Pos. 1)

Fig. 24: Removing the current sensor unit



- Pull out the entire current sensor assembly from the top of the GPA cabinet.
- Tilt the rack (like shown in the picture) to protect the cable to the CCU.

4.1.6 Installation

Proceed in reverse order of removal:

- Insert and secure the new current sensor assembly

CAUTION

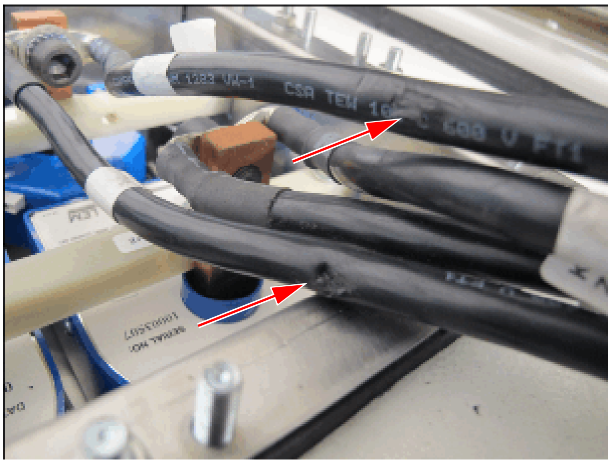
Potential damages of gradient cable insulation.

Risk of electrical shock!

» **Carefully check the gradient cables (visual control) for existing damages.**

» **In case of damage, repair the defect temporarily and replace the defective cable.**

Fig. 25: *Cable damage*



- Take care not to damage the insulation when fitting the cover over the cables.

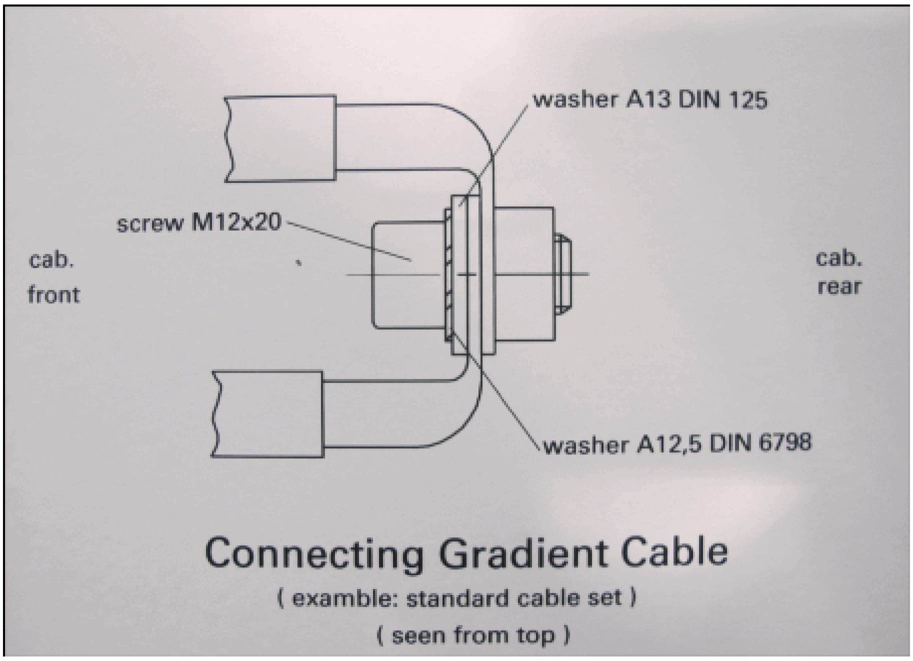
Fig. 26: *Heat shrink*

(1) Heat shrink

- In case of damaged insulation, repair the defect with heat shrink tube temporarily. (➔ Gradient Cables Outside the RF-Room / M6-015.812)
- Push the shrink tube over the cable from the RF-filter plate side.
- Shrink the tube with a heat gun.
- Replace the defect cable as soon as possible. (➔ Gradient Cables Outside the RF-Room / M6-015.812)

Ensure that the screws, washers and nuts are assembled in the correct sequence. See also the sticker at cabinet door.

Fig. 27: Connecting gradient cable



- Re-install the ground cable (only for shielded cable)

> The cable clamps must fasten the shield only.
> Ensure that the insulation is not damaged.

- After locking the screws at GPA roof- top, remove any residual metal dust from the current sensors.
- Install the GPA cover
- Reinstall the cables at the bottom of the current sensor assembly.
- Fasten the screws with a torque wrench. **Torque wrench** 25 ± 1 Nm).
- Reinstall the GSSU and all cable connections.

4.1.7 Concluding Steps

- Reinstall the front cover of the GSSU.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.

- Switch-on the system

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC..**

4.1.8 Adjustments

Perform TuneUP:

- Regulator Adjust
- Table Adjustment
- Phantom Shim
- CTC
- ECC
- Grad Sens
- Grad Delay

4.1.9 Final Checks

Perform Quality Assurance:

- Image Orientation Check
- Regulator Check
- Shim Check
- CTC Check
- ECC Check
- Grad Sens Check
- Gradient Delay Check
- Calc Artifacts
- Spike Check
- Stability Check
- FatSat Check

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

5.1 Power Stages (PS) K2217

(Material number 106 63 308)

D32 + fuses potentially defect as well (see (→ Additional Fuse Check / Page 57))

5.1.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury.

» **Switch-off the gradient cabinet with circuit breakers. (Both GPA Power Stage supply F110/ mains box and GSSU power supply F6/ ACC)**

WARNING

High voltage is still present on the final stage output terminals!

Contact with voltage conducting parts leads to death or serious physical injury.

» **After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.**

Check the green LEDs behind the 6 holes in the PS front side (→ Fig. 29 Page 53). **If the LEDs are on, the capacitors are charged, i.e. the supply voltage 400 VDC from D32 is connected to the PS.** If the green LEDs behind a single hole are off during normal operation, the corresponding part of the Power Stage is probably defective.

LED X1 corresponds to the substage X1 of X-Power Stage, the lower 2 LEDs (e.g. X3) belong to the same substage.	LEDs at PS (X,Y,Z)					
	X1	X5	Y1	Y5	Z1	Z5
	|	|	|	|	|	|
	X2	X4	Y2	Y4	Z2	Z4
	|	|	|	|	|	|
	X3	X3	Y3	Y3	Z3	Z3

It is also possible that after a single PS breakdown one of the corresponding 5 fuses between T1(sec) and D32 is blown. Refer to (→ Additional Fuse Check / Page 57).

Proceed with care when disconnecting/reconnecting the water hoses. Take every possible precaution to keep water from dripping into the cabinet. Water destroys the PS. Use a towel to catch excess water inside the quick connect.

Always handle the Power Stages with care (new and removed stages); especially when putting it on its back, it is highly likely that a small internal filtering board can break.

5.1.2 Tools and Time required

5.1.2.1 Time

155 min.

5.1.2.2 Tools

- Set of standard tools
- Ground test meter
- SW 19 wrench

5.1.3 Preliminary

- Switch-off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

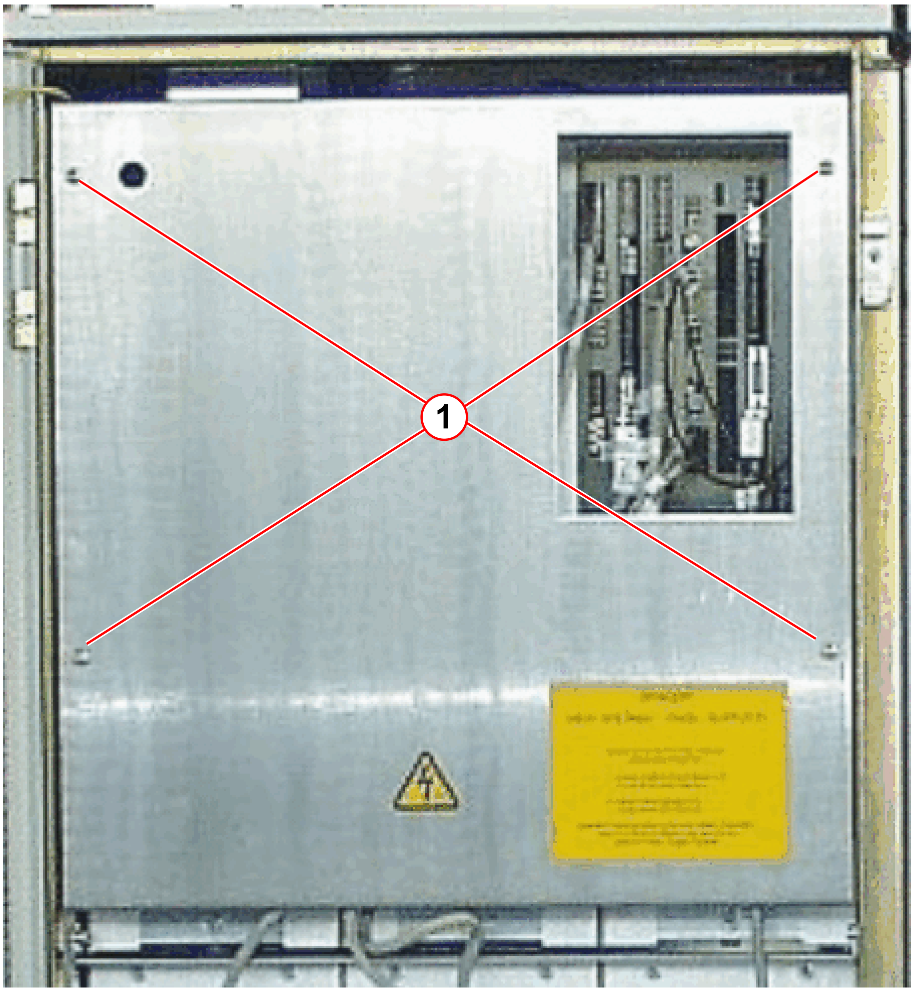
» **Adhere to the described steps for your own safety.**

- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

5.1.4 Disassembly

- Remove GSSU front cover (4 screws).

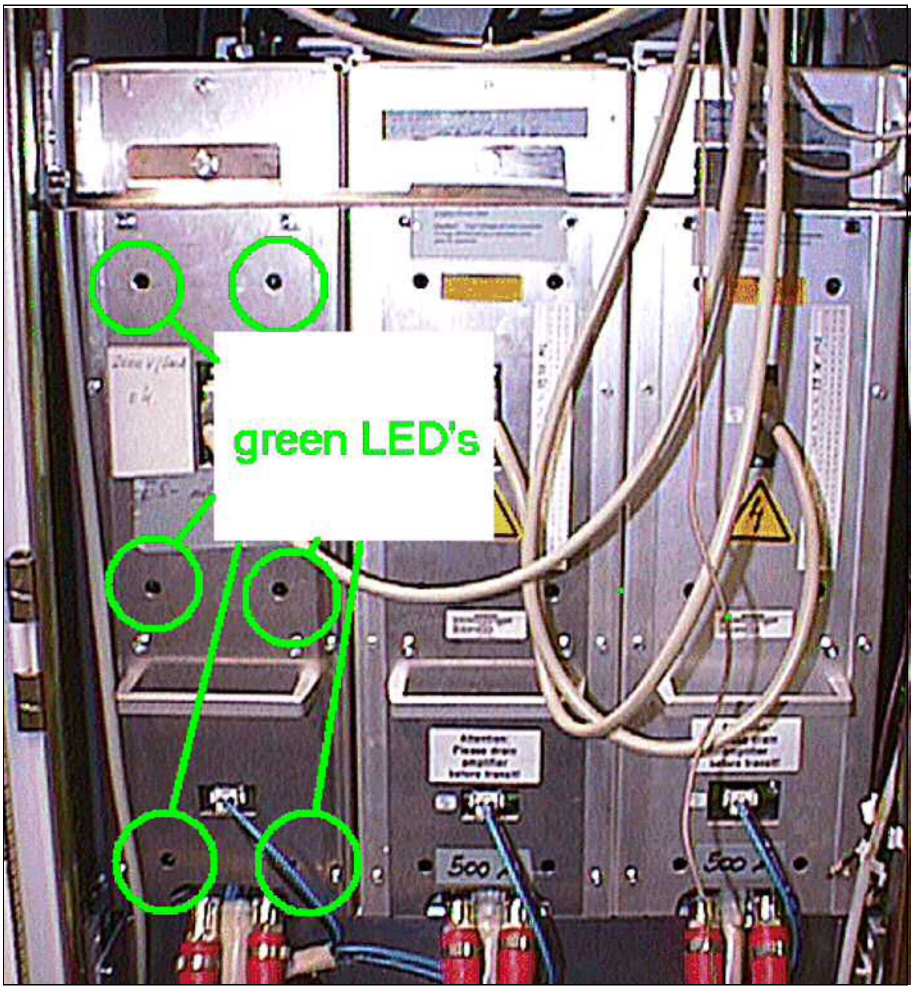
Fig. 28: GSSU - front cover



(1) Screws

Check the green LED's behind the 6 holes in the PS front side (→ Fig. 29 Page 53). If the LED's are on, the capacitors are charged! Later final stages may not have the LEDs fitted any more!

Fig. 29: View of GPA power stages



The arrangement of the power stages in the cabinet is as follows:

X - Y - Z.

5.1.4.1 Removing the Water Hoses

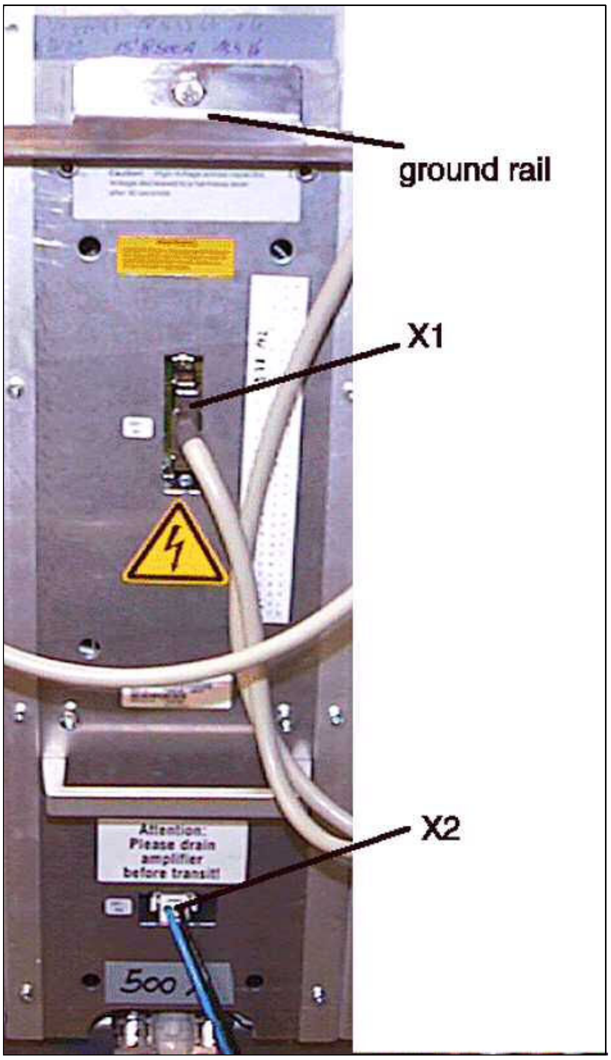
Fig. 30: Water distributor and water pipes to PS and Rectifier D32

- Remove the quick connections of the water hoses from the inlet and outlet sides (➔ Fig. 30 Page 54).

> **i** The water connections shut off automatically. However, ensure that there are no water spills and that splashes of water are cleaned up with a cloth.

5.1.4.2 Removing the Connectors and Cabling at PS and D32

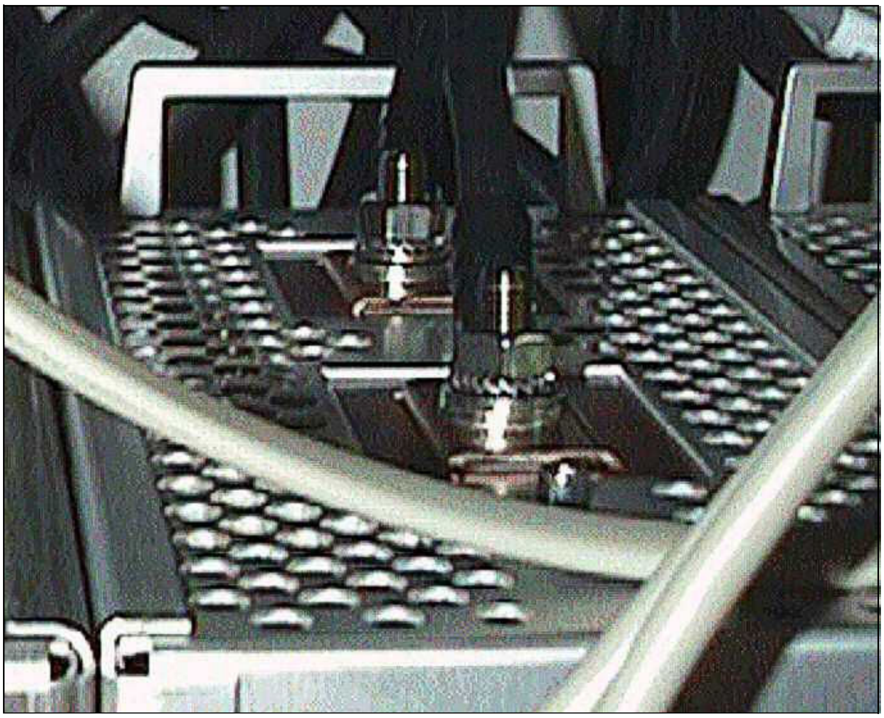
Fig. 31: Connectors at PS



- Disconnect the corresponding DCV supply connectors X12, 22 or 32 from D32.
- Disconnect the driver supply connector X2 from the Power Stage (→ Fig. 31 Page 55).
- Disconnect the 50-pin connector X1 from the Power Stage(→ Fig. 31 Page 55) .
- Loosen the gradient cables on top of the Power Stage (open-end wrench SW 19) (→ Fig. 32 Page 56).

5.1.4.3 Removing the Power Stage

Fig. 32: Power cable connection on top of power stage



- Remove the ground rail across the 3 Power Stages (➔ Fig. 31 Page 55).
- Remove the Power Stage using the handles on the top and front. (The weight of the PS is approx. 30 kg).

> **i** Do not drop off the final stage on the hoses to prevent the water connections from being damaged!

5.1.5 Assembly

Fig. 33: All power stages removed

- Slide in the replacement Power Stage. The guide cones shown in (→ Fig. 33 Page 57), engaged in a hole at the back of the PS, hold the Power Stage in position.
- Attach the securing rail across the 3 power stages (→ Fig. 31 Page 55).

> To prevent damage to the Power Stage by faulty control of the IGBT switches, check the SCSI-cable from D14 to the Power Stage for damage and the connectors for broken pins. Replace defective cables!

- Reconnect the gradient cables and all other connectors.
- Reconnect the DCV supply connector at D32.
- Connect the water hoses of the Power Stage with the water distributor as shown in (→ Fig. 30 Page 54). The longer, left water hose of the Power Stage is connected with the inlet of the water distributor.

5.1.5.1 Additional Fuse Check

If one or more power stages blow off, it is very likely that at least one of the corresponding 15 fuses (25A slow) between T1(sec) and D32 has been blown and/or that the D32 rectifier board is damaged on the corresponding axis.

- Access the fuse holders by removing the protection cover in front of D32, shown in (→ Fig. 35 Page 61).
- Pull the 5 fuse holders from the corresponding PS and check the fuses.

  Replace defective fuses (material number: 30 77 476).

  The arrangement of fuses is from left to right:

  X1...X5, Y1...Y5, Z1...Z5 (according to the partial power stages) (→ Power Stages (PS) K2217 / Page 50).

- In case of blown fuses, replace the D32 as well (➔ Rectifier D32 K2217 / Page 60).

Fig. 34: Fuse holders above D32 (behind the cover)

5.1.6 Concluding Steps

- Reinstall the front cover of the GSSU.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.
- Refill the secondary water circuit at the RCA with the drained water. Please refer to the **CB-DOC / Installation / Start-up** description (➔ Filling the secondary water circuit / M6-015.815.01).
- Switch on the water pump via the circuit breaker F1 in the RCA cabinet.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC. Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Switch on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

Before returning the replaced Power Stage, drain the water from the pipes by opening both ends.

5.1.7 Adjustments

Perform TuneUp:

- Regulator Adjust
- Phantom Shim
- CTC
- ECC
- Gradient Sensitivity
- Gradient Delay

Time 15 min.

5.1.8 Final Checks

Perform Quality Assurance:

- All selected procedures
- Image Orientation Check

Time 68 min.

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

6.1 Rectifier D32 K2217

(Material number 83 64 197)

6.1.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

6.1.2 Tools and Time required

6.1.2.1 Time

50 min.

6.1.2.2 Tools

- Standard tool set

6.1.3 Preliminary

- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

6.1.4 Disassembly

- Disconnect all quick connectors of water hoses from the water distributor.

i The water connections shut-off automatically. However, ensure that there are no water spills and that splashes of water are cleaned up with a cloth.

- Disconnect the DCV supply connectors to power stages X12, X22 and X32 from D32 (lower row of connectors).

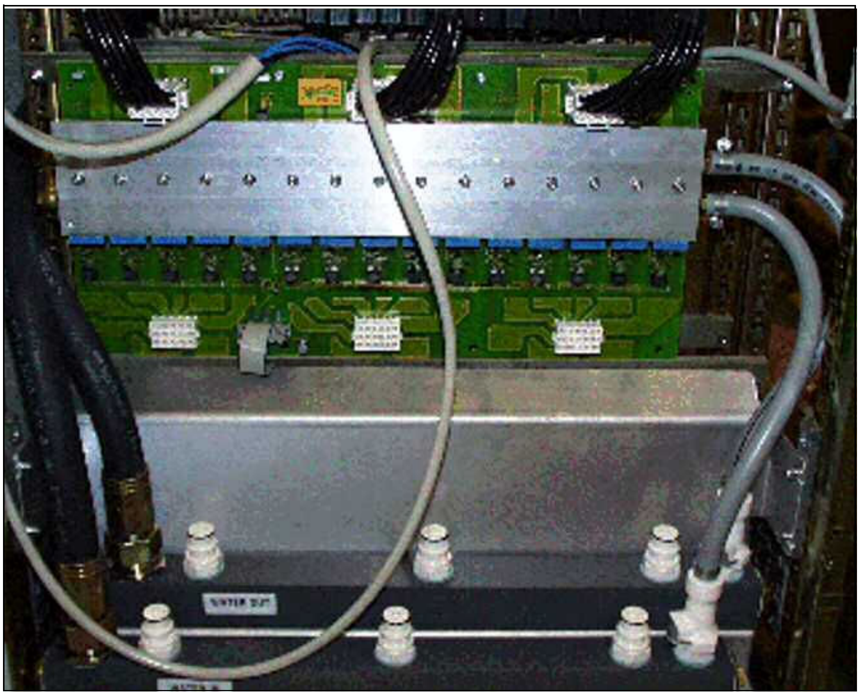
- Disassemble the protection cover by loosing the two nuts shown in (➔ Fig. 35 Page 61).

Fig. 35: Lower part of the GPA with protecting plates in front of D32

- Disconnect the AC-supply connectors X 11, X21 and X31 (upper row of connectors of D32).

- Remove flat cable connector X1 to D31 from D32.

Fig. 36: View of D32 with removed protecting plate



- Loosen the four securing screws/nuts at the upper edge of D32.
- Remove the rectifier board D32.

6.1.5 Assembly

- For the installation of D32, proceed in reverse order.

- Install the protection cover by loosing the two nuts shown in (→ Fig. 37 Page 63).

Fig. 37: Lower part of the GPA with protecting plates in front of D32

6.1.6 Concluding Steps

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» First switch on circuit breaker F110 / mains box, then F6/ ACC

- Switch-on the water pump via the circuit breaker F1 in the RCA cabinet.
- Switch-on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

Before returning the replaced rectifier board D32, drain the water from the pipes by opening both ends.

6.1.7 Final Checks

Measure an MR image

Perform Quality Assurance:

- Regulator Check

Time 10 min.

6.2 Soft Start D31 K2217

(Material number 57 26 414)

6.2.1 Safety Information

WARNING

High Voltage

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

6.2.2 Tools and Time required

6.2.2.1 Time

60 min.

6.2.2.2 Tools

- Standard tool set

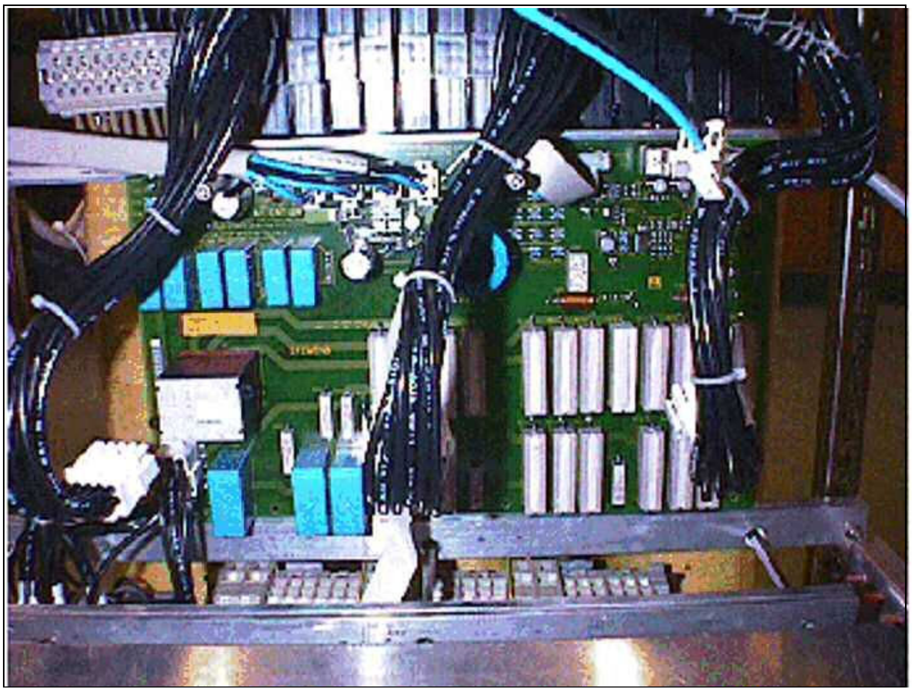
6.2.3 Preliminary

- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

6.2.4 Disassembly

- Remove D32 first (→ Rectifier D32 K2217 / Page 60).
- Remove connector X1 to GSSU from D31.
- Remove connectors X10 to GSSU power supply from D31.
- Remove connectors X11,12,13 to power stages from D31.
- Remove flat cable connector X8 to D32 from D31.
- Remove connector X2 to main contactor K2 from D31.
- Loosen the four securing screws/nuts at the upper edge of D31.
- Remove D31.

Fig. 38: View of D31 (D32 removed)



6.2.5 Assembly

- For the installation of D31, proceed in reverse order.

- Install the protection cover by loosing the two nuts shown in (➔ Fig. 39 Page 67).

Fig. 39: Lower part of the GPA with protecting plates in front of D32

6.2.6 Concluding Steps

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» First switch on circuit breaker F110 / mains box, then F6/ ACC

- Switch on the water pump via the circuit breaker F1 in the RCA cabinet.
- Switch on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

Before returning the replaced rectifier board D32, drain the water from the pipes by opening both ends.

6.2.7 Final Checks

Measure an MR Image

Perform Quality Assurance:

- Regulator Check

Time 10 min.

6.3 Main Contactor K2 GPA K2217

(Material number 30 90 560)

6.3.1 Safety Information

WARNING

High Voltage

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

6.3.2 Tools and Time required

6.3.2.1 Time

80 min.

6.3.2.2 Tools

- Standard tool set
- Torx 25 driver bit

6.3.3 Preliminary

- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.

6.3.4 Disassembly

The following description refers to the worst case, i.e., the left side of the GPA cabinet is not accessible.

If the GPA cabinet is free-standing in the computer room, the replacement can be simplified considerably by removing the left side panel of the gradient cabinet. Then you can remove the mounting plate together with the K2 by loosening the two Torx 25 screws (→ Fig. 42 Page 72).

- Remove D32 first (→ Rectifier D32 K2217 / Page 60).
- Remove the water distributor (→ Water Distributor GPA K2217 / Page 87). The water hoses can remain connected.
- Remove connector X2 from D31 (→ Fig. 41 Page 71).
- Loosen the screws (3mm Allen), which secure the K2 on the mounting plate (→ Fig. 43 Page 73) and (→ Fig. 44 Page 74).

- Remove the power cables from K2 (to transformer T1, bottom, and line filter, top).
- Remove the cables from K2 to X2/D31.
- Remove K2.

Fig. 40: View from open left side of GPA cabinet of D31, D32 and main contactor K2

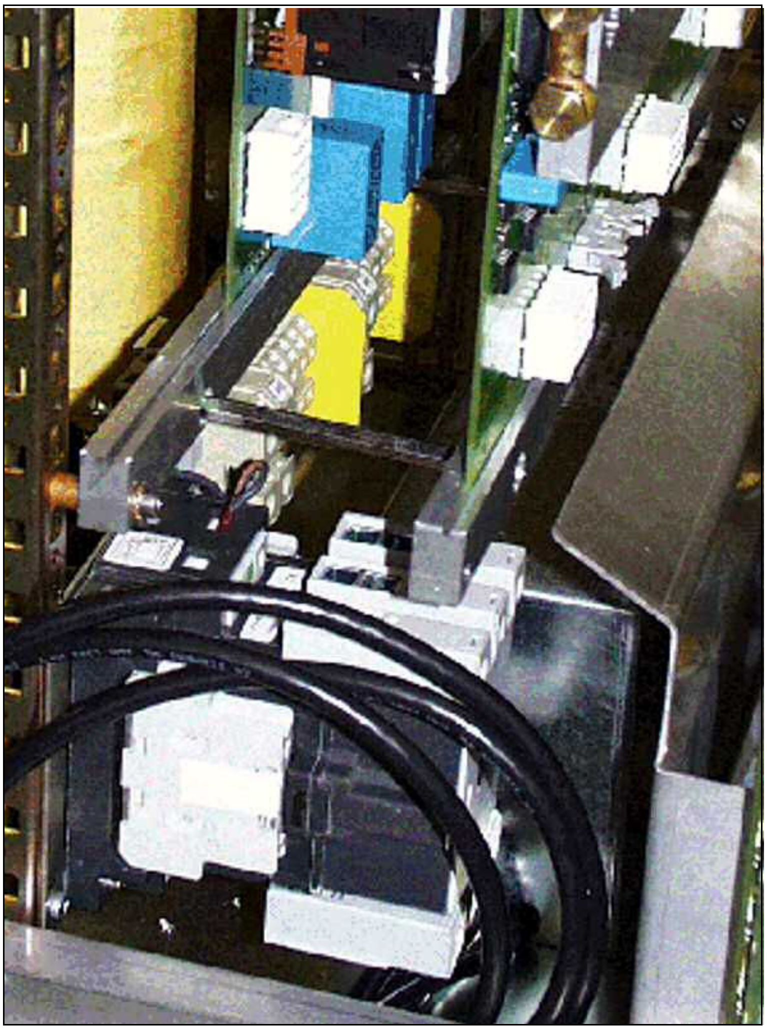


Fig. 41: Connector X2 on D31 and main contactor K2

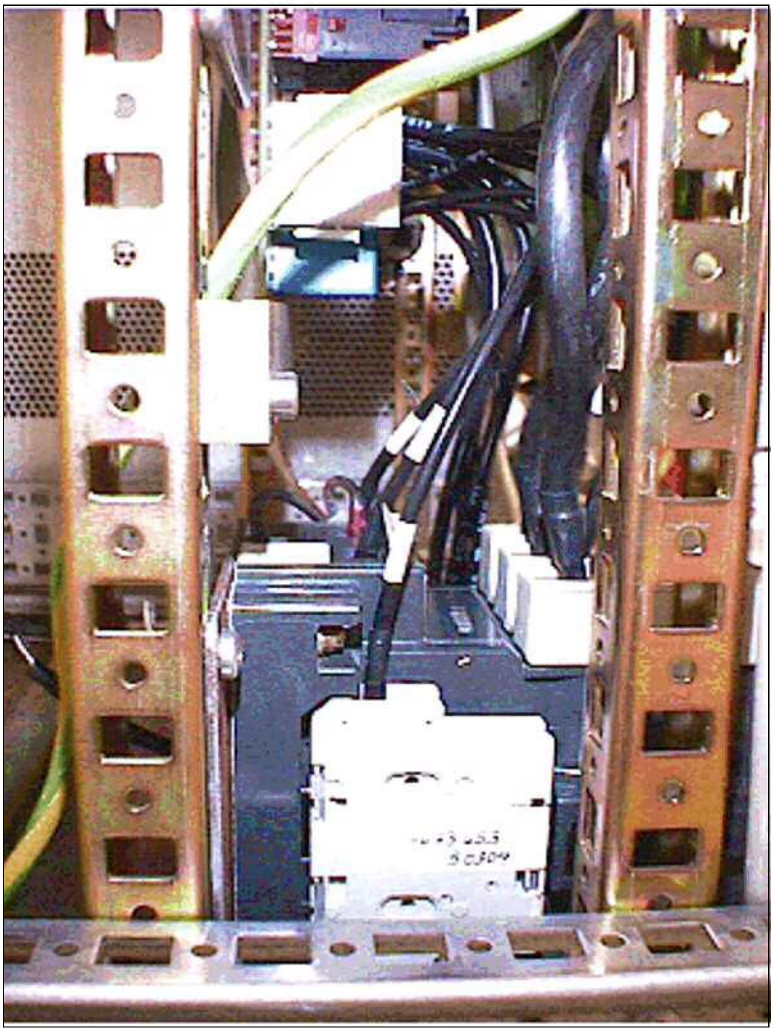


Fig. 42: Mounting plate of K2 secured with two Torx 25 screws

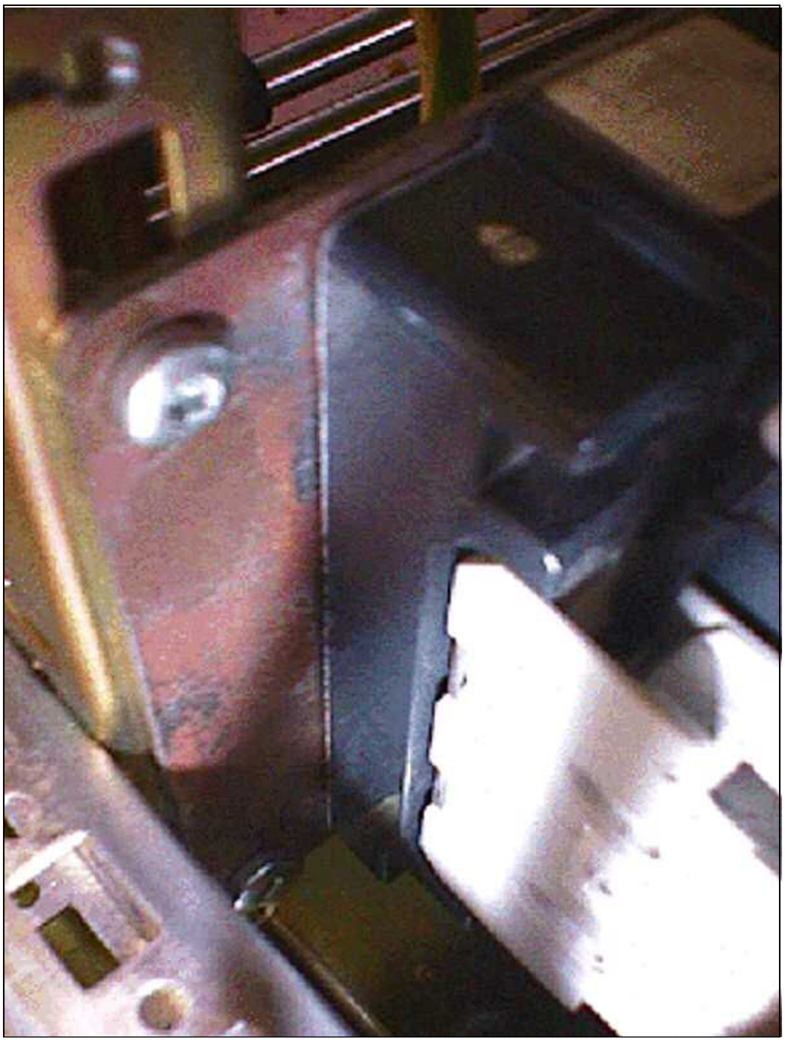


Fig. 43: Stabilizing screw (3 mm Allen) on lower left of main contactor

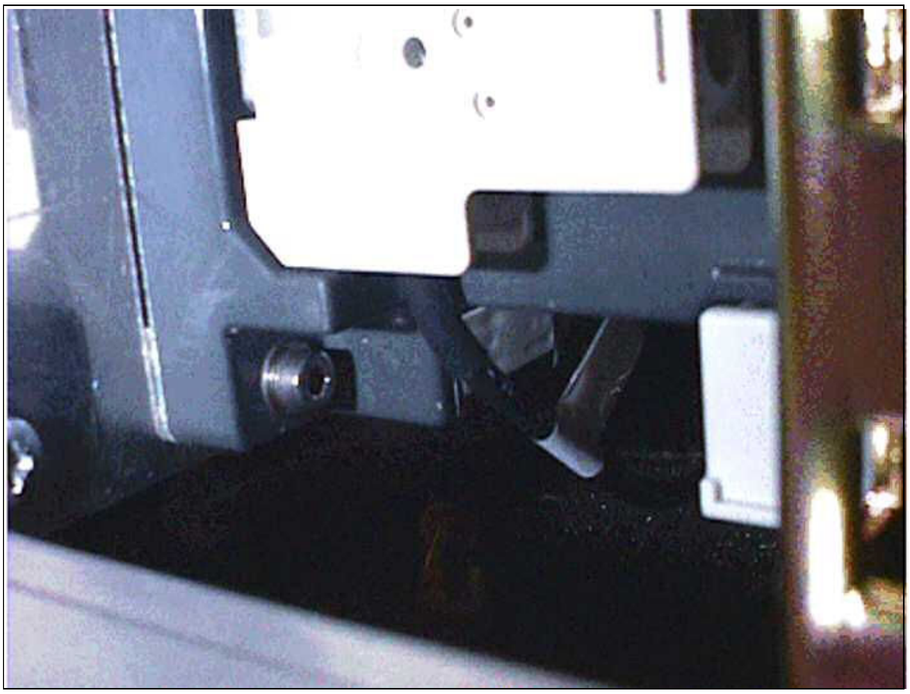
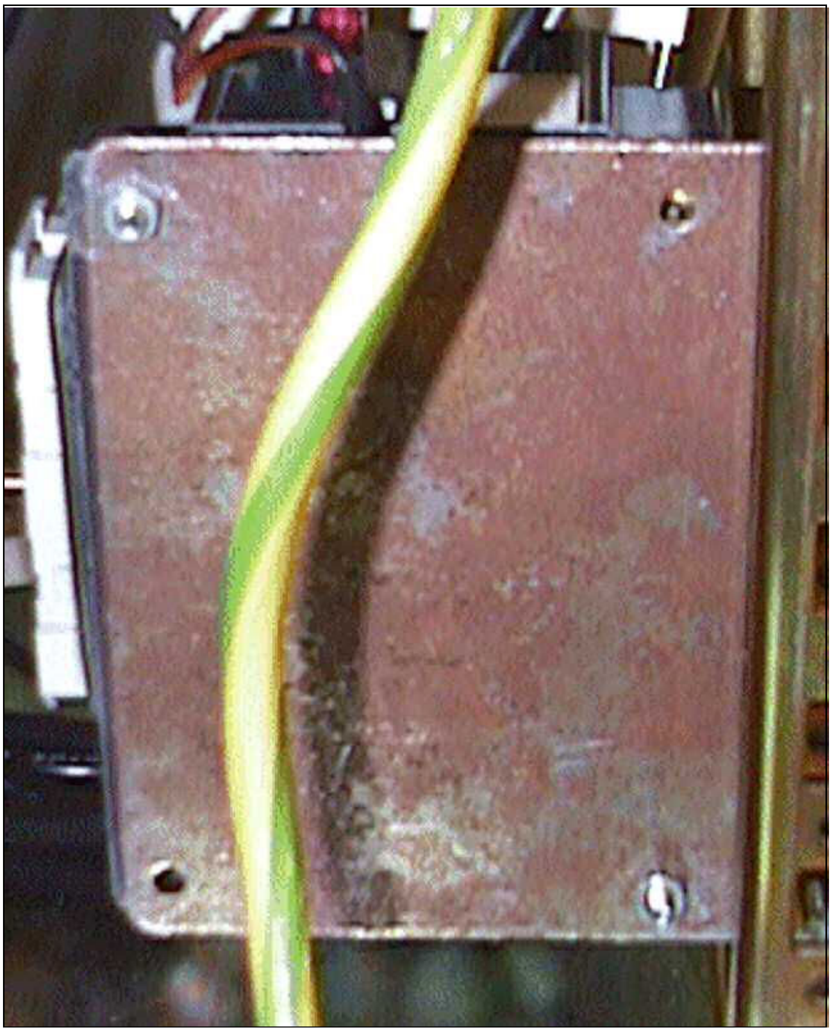


Fig. 44: Back of K2 mounting plate



6.3.5 Assembly

- To install the line filter, proceed in reverse order.

- Install the protection cover by loosing the two nuts shown in (➔ Fig. 45 Page 75).

Fig. 45: Lower part of the GPA with protecting plates in front of D32

6.3.6 Concluding Steps

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Switch-on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

6.3.7 Final Checks

- Measure an MR image.

Time 5 min.

7.1 Chokes of GPA K2217

(Material number 57 25 887)

7.1.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

CAUTION

High Voltage across capacitor/gradient cable connections.

Risk of electrical shock!

» **After turning off the GPA cabinet, the voltage decreases to a safe level after 20 minutes.**

CAUTION

Hot components

Particularly after a heavy load, the output filter chokes, the transformer and rectifier circuits can get very hot. Touching these parts can cause severe skin burn.

» **Do not touch these parts until at least 30 min. after you have shut down the GPA.**

7.1.2 Tools and Time required

7.1.2.1 Time

190 min.

7.1.2.2 Tools

- Standard tool set
- Ground test meter

7.1.3 Preliminary

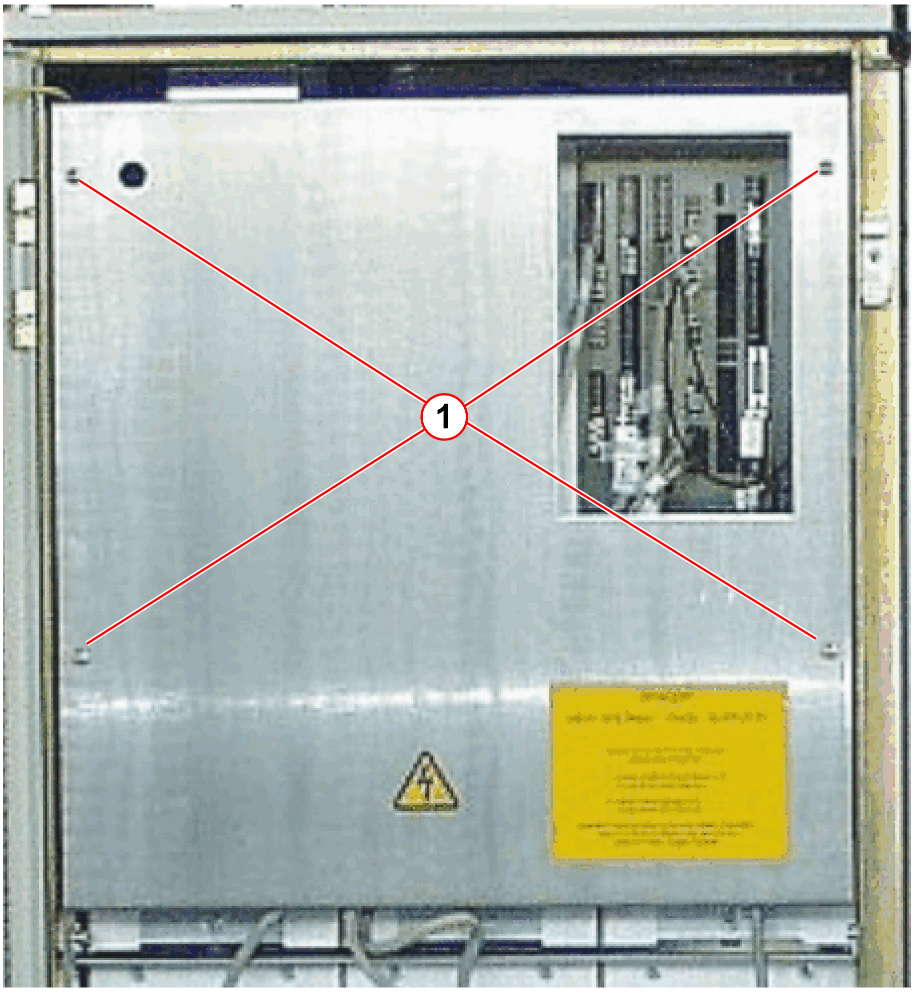
Always replace the 2 chokes of one gradient axis circuit together.
Replace the corresponding Power Stage as well.

- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.
- Switch off the water pump via the circuit breaker F1 in the RCA cabinet.

7.1.4 Disassembly

- Remove GSSU front cover (4 screws).

Fig. 46: GSSU - front cover



(1) Screws

- Remove all Power Stages (→ Power Stages (PS) K2217 / Page 50).
- For disconnecting the cables from the current sensors, refer to (→ Disconnecting the Gradient Cables in the GPA / Page 44)

- Loosen the screws which hold the supporting rails (→ Fig. 47 Page 79).

Fig. 47: The six chokes in the back of GPA cabinet - PSs are removed

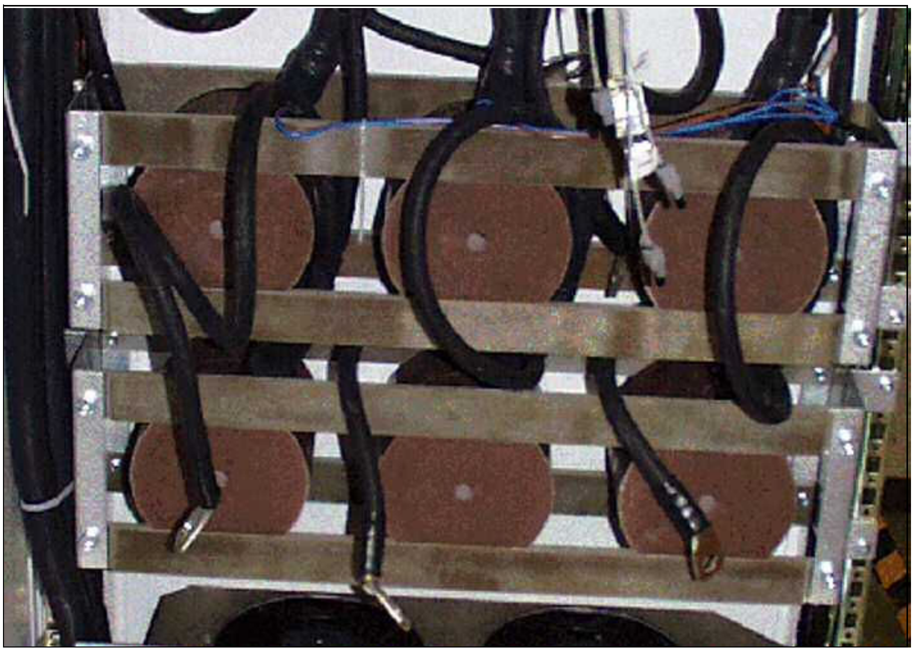


Fig. 48: Temperature switch at one of the upper chokes

- Remove the relevant chokes.

> The upper 3 chokes might have built-in thermo switches. If your system is equipped with temperature switches, disconnect the cables from the connectors (→ Fig. 48 Page 79).

7.1.5 Assembly

- For the installation of the chokes proceed in reversed order.

- Install the GSSU front cover.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.

7.1.6 Concluding Steps

- Switch on the water pump via the circuit breaker F1 in the RCA cabinet.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Switch-on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

7.1.7 Adjustments

Perform TuneUp:

- Regulator Adjust
- Phantom Shim
- CTC
- ECC
- Gradient Sensitivity
- Gradient Delay

50 min.

7.1.8 Final Checks

Check the function of the blowers after a choke was exchanged.

Perform Quality Assurance:

- All selected procedures
- Image Orientation Check
- Time 68 min.

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

7.2 Blowers of GPA K2217

(Material number 30 63 476)

7.2.1 Safety Information

WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

CAUTION

High Voltage across capacitor/gradient cable connections.

Risk of electrical shock!

» **After turning off the GPA cabinet, the voltage decreases to a safe level after 20 minutes.**

CAUTION

Hot components

Particularly after a heavy load, the output filter chokes, the transformer and rectifier circuits can get very hot. Touching these parts can cause severe skin burn.

» **Do not touch these parts until at least 30 min. after you have shut down the GPA.**

i Proceed with care when disconnecting/reconnecting the water hoses. Take every possible precaution to keep water from dripping into the cabinet. Water destroys the PS. Use a towel to catch excess water inside the quick connectors.

7.2.2 Tools and Time required

7.2.2.1 Time

115 min.

7.2.2.2 Tools

- Standard tool set

- Ground test meter

7.2.3 Preliminary

All 3 Power Stages must be removed to replace the GPA blowers.

- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.

WARNING

High Voltage

Contact with voltage conducting parts leads to death or serious physical injury!

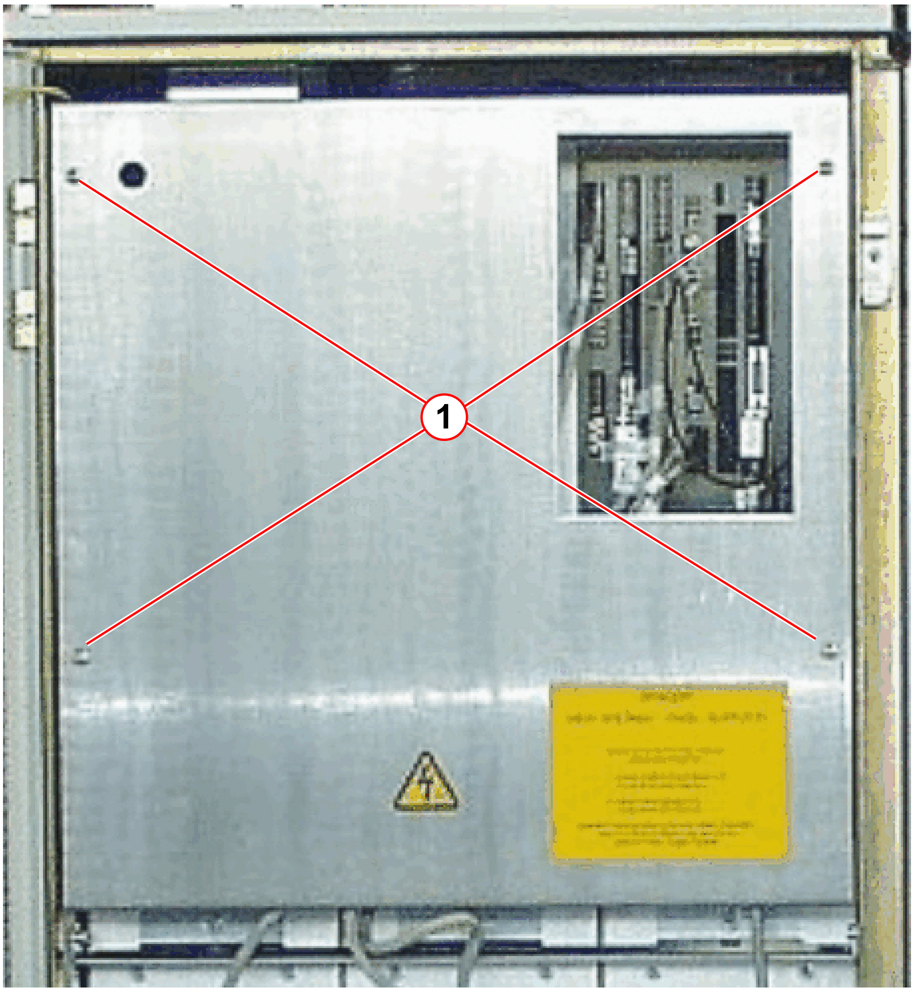
» **Adhere to the described steps for your own safety.**

- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

7.2.4 Disassembly

- Remove GSSU front cover (4 screws).

Fig. 49: GSSU - front cover

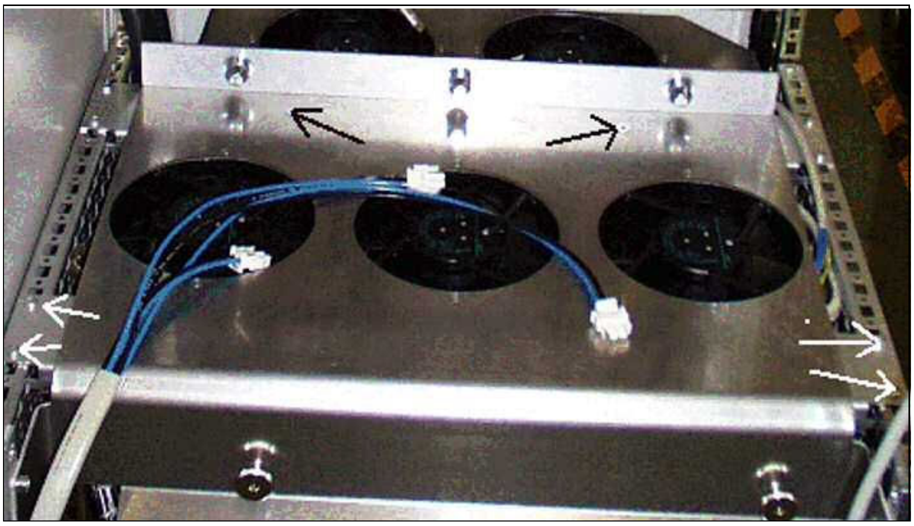


(1) Screws

- Remove all 3 power stages (→ Power Stages (PS) K2217 / Page 50).
- Loosen the 2 stabilizing screws at the right and left side of the blower mounting plate and the 2 screws in back of the blower mounting plate, shown in (→ Fig. 50 Page 85).
- Remove the mounting plate.

- Disconnect the cables from the blower connectors.

Fig. 50: Blower assembly (five blowers)



7.2.5 Assembly

- For the installation of the blowers, proceed in reverse order.

7.2.6 Concluding Steps

- Install the GSSU front cover.
- Measure the protective earth connection resistance <= 100 mOhm with a ground test meter between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Switch on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).
- Check the proper function of all blowers.

7.2.7 Final Checks

Measure an MR Image

Perform Quality Assurance:

- Image Orientation Check

Time 10 min.

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

8.1 Water Distributor GPA K2217

(Material number 73 70 146)

8.1.1 Safety Information

⚠ WARNING

High Voltage!

Contact with voltage conducting parts leads to death or serious physical injury!

» **Adhere to the described steps for your own safety.**

8.1.2 Tools and Time required

8.1.2.1 Time

195 min.

8.1.2.2 Tools

Standard tool set

8.1.3 Preliminary

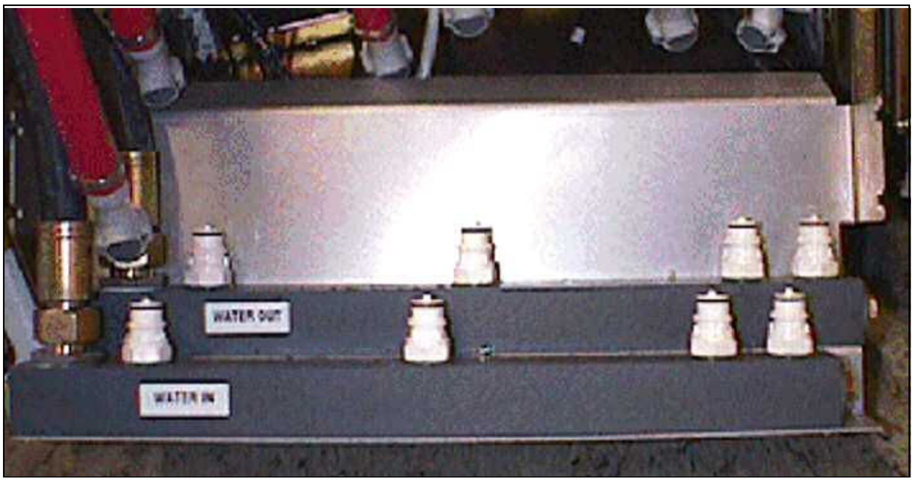
- Switch off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

8.1.4 Disassembly

- Disconnect all quick connectors of water hoses from the water distributor.
- Loosen the 2 screws which hold the cover plate on the GPA cabinet frame at the left and right side (→ Fig. 51 Page 88).
- Take the water distributor out of the GPA cabinet (→ Fig. 52 Page 88).
- To empty the water circuit, carefully open the lateral seals on the right side of the distributors and collect water in a bucket (first at inlet side, then outlet side).
- Remove the 2 water connections for the inlet and outlet (black water hoses).
- Refill the secondary water cooling circuit in the RCA with the drained water.
- Check the cooling water connections for leaks.

Fig. 51: Water distributor - all hoses are connected

Fig. 52: Disassembled water distributor



8.1.5 Assembly

- For the installation of the water distributor, proceed in reversed order.
- Refill the secondary water circuit at the RCA with the drained water. Please refer to the **CB-DOC/Installation/Start-up** description (➔ Filling the secondary water circuit / M6-015.815.01).

8.1.6 Concluding Steps

- Switch on the water pump via the circuit breaker F1 in the RCA cabinet.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Switch on circuit breaker and F110 on top of the ACC and then F6 (in the ACC).

8.1.7 Final Checks

- Check the cooling water connections for leaks.
- Measure an MR image.

Time: 5 min.

9.1 GC Water Distributor

(Material number 59 38 688)

9.1.1 Safety Information

WARNING

Strong magnetic field near the gradient coil water connections exerts high attraction on magnetic objects.

Attracted objects can lead to serious physical injuries or death!

» Use non-magnetic tools. If not available, the magnet has to be ramped down.

WARNING

High Voltage

Contact with voltage conducting parts leads to death or serious physical injury!

» Adhere to the described steps for your own safety.

9.1.2 Tools and Time required

9.1.2.1 Time

150 min.

9.1.2.2 Tools

- Non-magnetic standard tool kit (77 57 706)
- Non-magnetic scalpel

9.1.3 Preliminary

- Switch-off circuit breaker F6 (in the ACC) and F110 at top of ACC.
- Wait 20 minutes until the capacitors in the power stages are fully discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

9.1.4 Disassembly

9.1.4.1 Disassembling the Magnet Covers

To access the water distributors, disassemble the back cover/ funnel of the magnet. For details of this procedure, refer to the **CB-DOC/Installation Instructions/Installation of Covers**.

9.1.4.2 Draining the Water Circuit of the Gradient Coil

- Disconnect the quick coupling of the water distributors (inlet and outlet flow).
- Mark and note the correct positions of the connections at the gradient coil on the replacement hoses. (The hoses are already connected to the replacement water distributor).
- Cut the water hoses to the gradient coil at a length of approximately 15 cm from the distributor. Have plastic buckets prepared to collect the cooling fluid/ water.
- Wait until cooling fluid/ water is fully drained.

9.1.4.3 Disassembling the Water Hose from the Nozzle

Examples of hose connections at water distributors:

Fig. 53: Hose connection to water distributor

Fig. 54: Detail hose nozzle - water distributor

Examples of hose connections at gradient coils:

Fig. 55: Example 1 - water hose connection at gradient coil

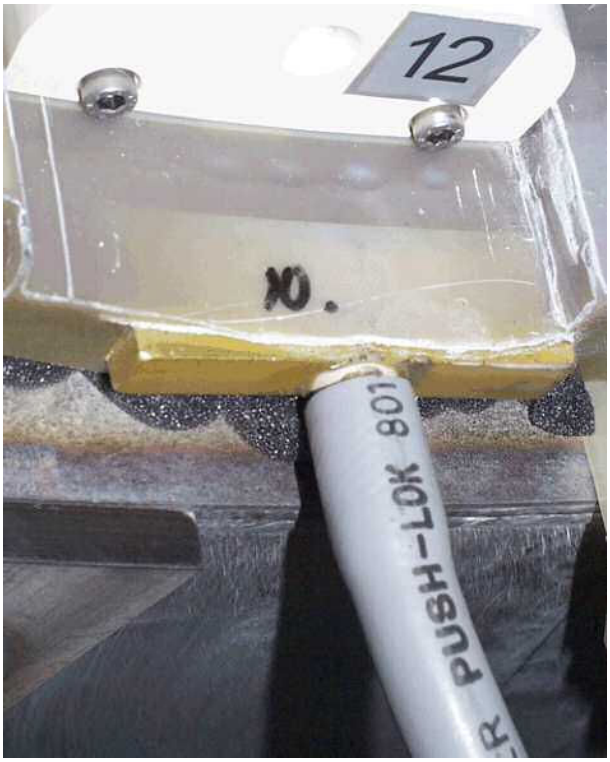


Fig. 56: Example 2 - water hose connection at gradient coil

Fig. 57: Example - hose nozzles at gradient coil

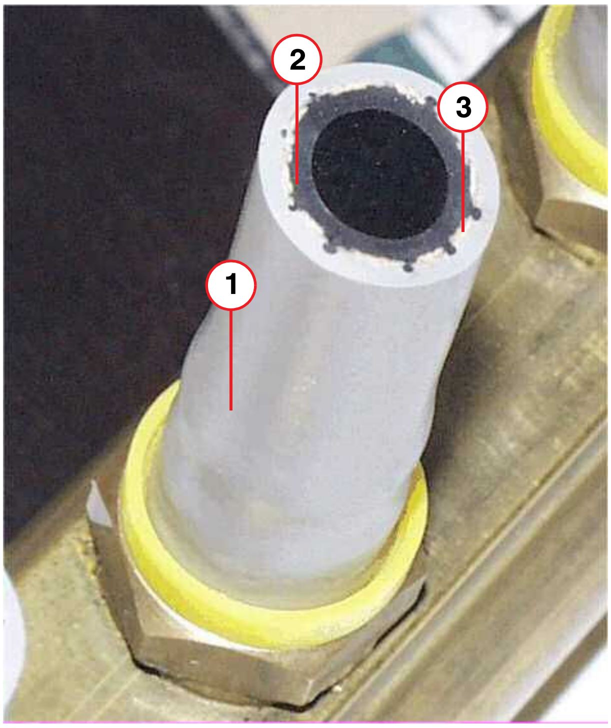
Structure and property of the water hose

The water hose is made of 3 layers (→ 1,2,3/Fig. 58 Page 93):

- the outer grey upper layer (1)
- the white web layer (3) and
- the black inner rubber layer (2).

The web (3) has the property to contract if the hose is pulled from the nozzle. Because the web is not flexible, the hose is also contracted and is fastened on the nozzle. By pulling, the hose cannot be removed from the nozzle. Otherwise, during pushing the hose onto the nozzle, the web widens which facilitates pushing the hose over the nozzle.

Fig. 58: Structure of water hose



(1) Grey rubber layer
(2) Black rubber layer
(3) White web layer

Removing the hose from the nozzle

NOTICE

To remove the water hose from the nozzle, never cut perpendicularly to the surface of the hose and parallel to the brass nozzle (➔ Fig. 59 Page 94).

If the nozzle surface becomes scratched, the connection will never be tight, and in the worst case the gradient coil has to be replaced.

» Scrape parallel to the hose surface. Remove the grey outer rubber layer, disrupt the white web, but do not completely remove the black inner rubber layer.

- Scrape carefully with a non-magnetic scalpel parallel to the surface of the hose.
- Cut through the web, but do not scrape too much of the underlying black rubber layer (➔ Fig. 60 Page 94).

Fig. 59: Never cut the hose perpendicular to the hose and parallel to the hose nozzle

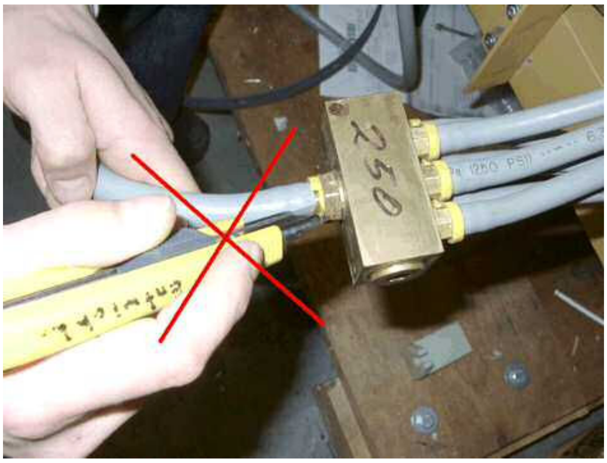


Fig. 60: Scraping the water hose at the hose nozzle

- If the web is destroyed, the hose can be removed from the nozzle by manually bending it considerably ,(➔ Fig. 61 Page 94), (➔ Fig. 62 Page 94).

Fig. 61: Removing the hose from the nozzle

Fig. 62: Scraping parallel to hose surface

9.1.5 Assembly

- Wet the nozzle with water.
- Push the hose applying appropriate force totally across the nozzle.

9.1.6 Concluding Steps

- Refill the secondary water circuit at the RCA with the drained water. Please refer to the **CB-DOC/Installation/Start-up** description (➔ Filling the secondary water circuit / M6-015.815.01).
- Check tightness of hose connections and water distributor.

- Reassemble the magnet covers.

NOTICE

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F110/ mains box MUST be switched on BEFORE powering up the GSSU with F6/ ACC.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

» **First switch on circuit breaker F110 / mains box, then F6/ ACC.**

- Switch-on circuit breaker F110 on top of the ACC and then F6 (in the ACC).

9.1.7 Final Checks

- Measure an MR image.

Time: 5 min.

10.1 Replacing the Gradient Cable Kit

10.1.1 Safety Information

WARNING

High Voltage!

Contact with voltage-conducting parts leads to death or serious physical injury!

» **Switch off the gradient cabinet**

» **Gradient cabinet K2217: Cutter switch F110 in the ACC mains box and F6/ACC must be switched off.**

WARNING

High voltage is still present on the final stage output terminals!!!

Contact with voltage-conducting parts leads to death or serious physical injury!

» **After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.**

Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

10.1.2 Tools and Time required

The new gradient cable kit 11 41 923 (between the Gradient Coil and the magnet connector panel) uses the Sonata cables (50 mm^2 for all systems) .

10.1.2.1 Time

180 min.

10.1.2.2 Contents of the Cable Repair Kit

Replacement kit 11 41 923 contains the parts listed in the table below.

Tab. 6 Contents of cable kit 11 41 923

Item	Part no.	Quantity	Description
1	5940502	1	Gradient cable kit Sonata 50 mm^2
2	5940577	12	Cable clamp 50 mm^2
3	3424793	12	Screw M4x30

Item	Part no.	Quantity	Description
4	3342052	12	Washer A4.3
5	3417631	12	Nut M4
6	3432911	6	Screw M10x20
7	7063969	13	Nord-Lock washer M10
8	7063761	6	Nord-Lock washer M12
9	5938670	1	Protective cover
10	3539236	1	Loctite_211

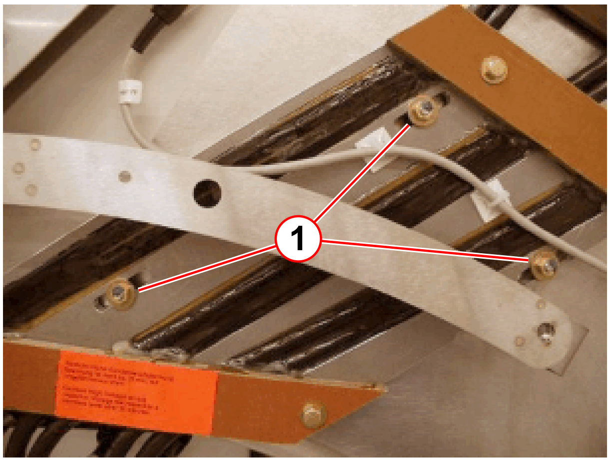
10.1.2.3 Tools and additional materials required

- Non-magnetic standard tool set (77 57 706)
- Non-magnetic torque wrench ranging from 5 - 20 Nm (83 95 829)
- Non-magnetic torque wrench ranging from 20 - 50 Nm (83 95 803)
- Non-magnetic socket assortment (83 96 116)

10.1.3 Procedure

- Remove the rear cover to gain access to the connections
- Remove 3 x M6 (10-mm open wrench) nuts to the magnet (➔ 1/Fig. 63 Page 97)
- Remove 6 x M12 (19-mm open wrench) nuts (➔ Fig. 64 Page 97)

Fig. 63: Gradient plate attachment



(1) 3 of the 10 mm nuts

Fig. 64: Connection plate

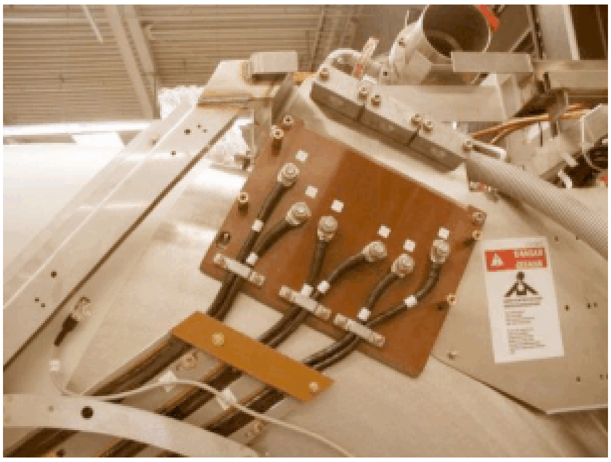


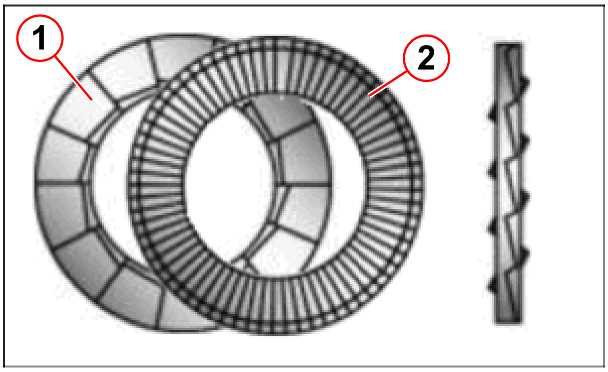
Fig. 65: Strain relief

- Remove 3 x metal strain relief clamps at the connection plate (➔ Fig. 65 Page 98).
- Remove the gradient plate.

- Replace the gradient plate with the delivered Sonata version (50 mm^2) and attach it to the magnet in reverse order using the 10 mm nuts (only hand-tighten them at first).

> Before connecting the gradient cables, check the cable insulation for defects to prevent short circuits. Make absolutely sure that the cables are not touching any conductive materials.

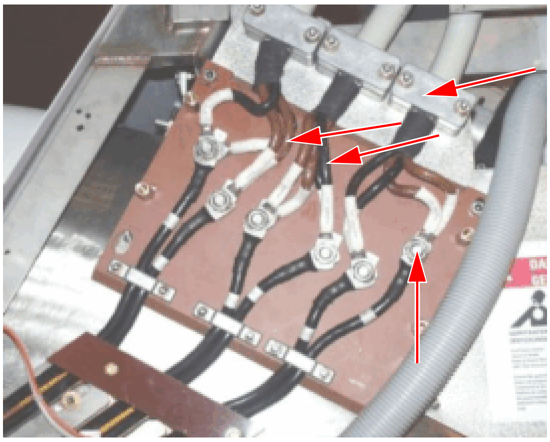
Fig. 66: NORD-LOCK washer



(1) Cam side
(2) Ridged side

A set of Nord-Lock washer must be placed between the cable lug and nut when fixing the cables at the gradient coil connection. The correct order of the washers (➔ Fig. 66 Page 98) is important for a proper securing function.

Fig. 67: Gradient connection plate



- Connect the gradient cables at the connection plate (→ Fig. 67 Page 99) using 6 of the Nord-Lock washer (M12) delivered with the kit.
- Check to ensure that the gradient cables are clamped correctly.
- Tighten the nuts with a non-magnetic torque wrench to

35 ± 2 Nm.

- Repeat the checks and have a second person confirm both torque and polarity.

- Check to see that the torque is  35 ± 2Nm.

- Mark the connections (nuts/bolts) with a waterproof pen to confirm they have been tightened with a torque wrench.
- Attach the cables at the gradient coil according to the wiring schematics (→ Fig. 68 Page 99) and use the M10 Nord-Lock washer in the correct order (→ Fig. 69 Page 99).

Fig. 68: Wiring at the gradient coil

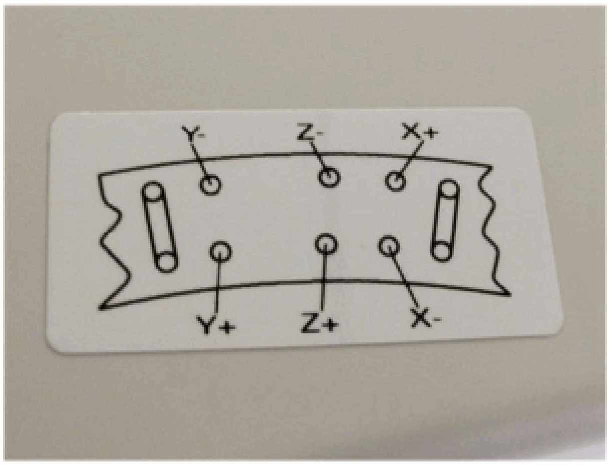
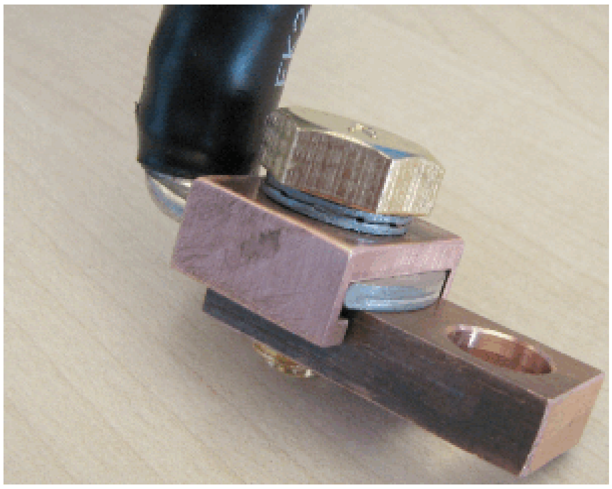


Fig. 69: Correct coil connection arrangement



Route the cables so that the lugs are even with the copper retaining bolts at the coil to achieve a good level of contact. The leads should be routed in parallel to each other (➔ Fig. 70 Page 100).

Ensure that the cable lugs do not come into contact while tightening the M10 x 20-hex brass screws.

Use the delivered NORD-LOCK washers contained in the repair kit on all M10x20 screws at the Gradient Coil (13 in total).

Fig. 70: Cables routed correctly

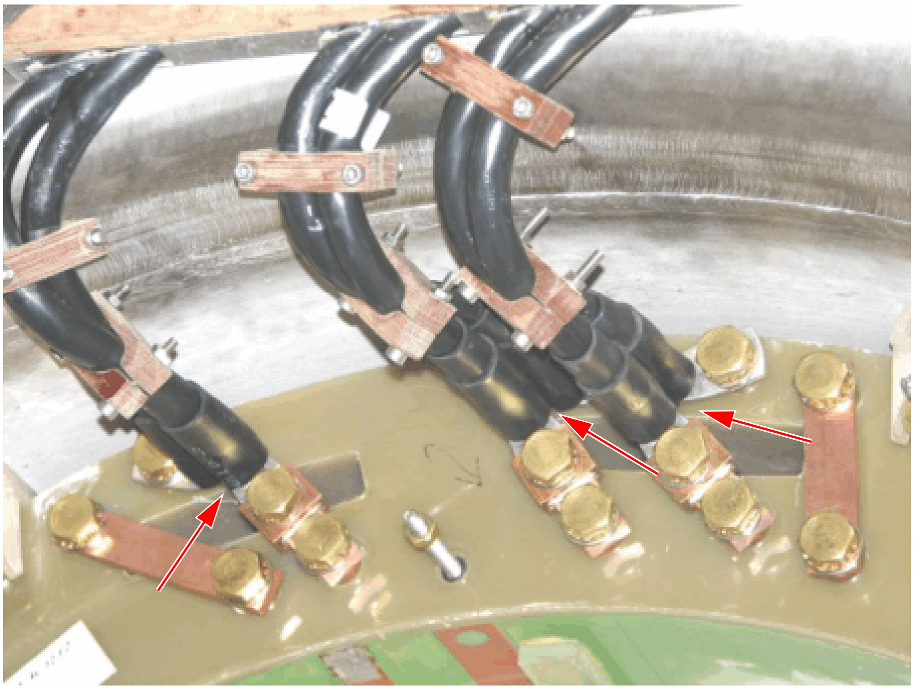
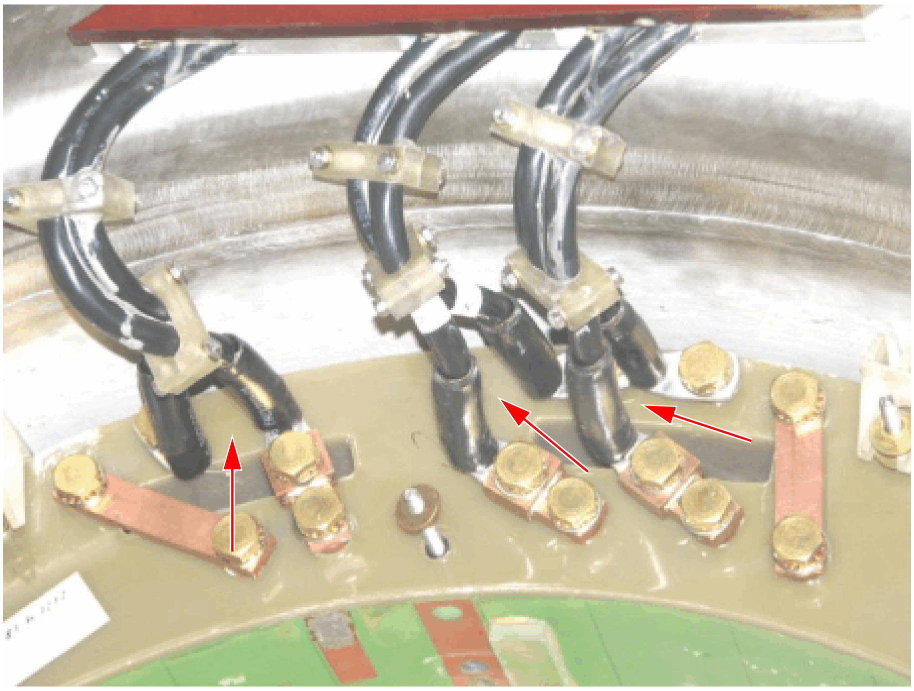


Fig. 71: Cables routed incorrectly



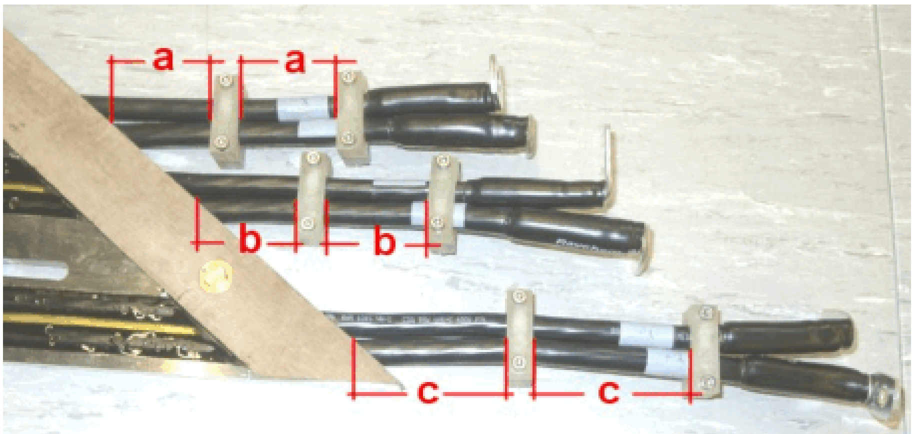
- Use a **non-magnetic** torque wrench to tighten the M10x20 hex brass screws at the gradient coil.

  »  **Torque 22 ± 1 Nm**

- Now fully tighten the 3 x M6 nuts holding the gradient plate in place (➔ Fig. 63 Page 97).
- Fasten the cable clamps, secure them with Loctite and tighten them to **5 Nm**.
  - » Fasten the first clamp as tight as possible to the heat shrink tubing (➔ Fig. 72 Page 102).
  - » Position the clamps equally along the length of the leads (➔ Fig. 72 Page 102).

Cable ties are not permitted.

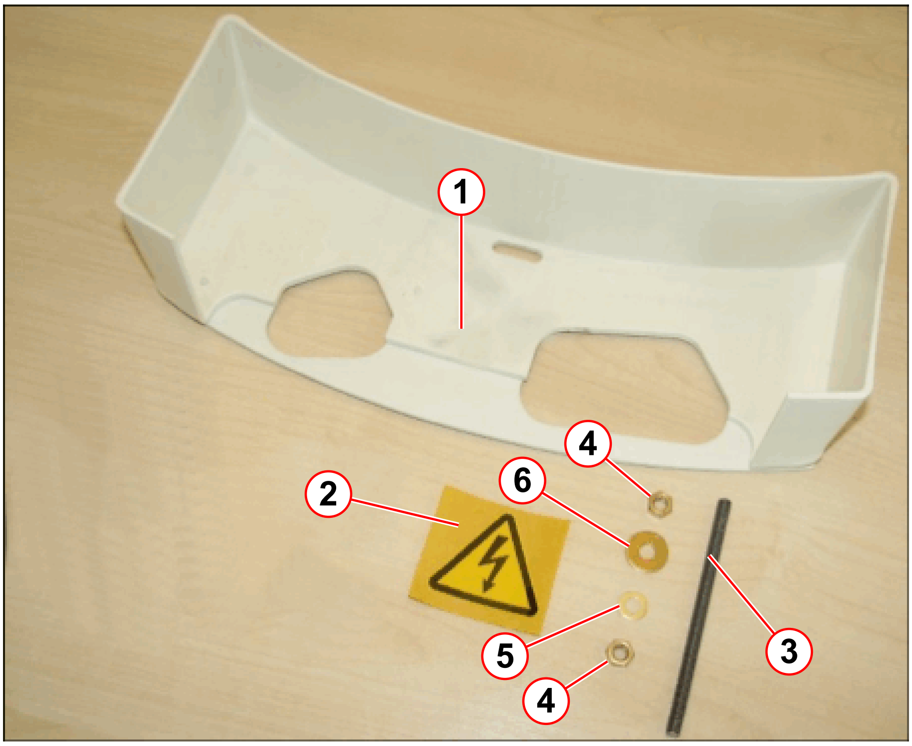
Fig. 72: Position of the cable clamps



10.1.3.1 New Protective Cover on Gradient Coil Terminals

If a previous model of gradient coil cover was installed, it must be replaced with the new version (→ 1/Fig. 73 Page 103) provided in the spare parts kit.

Fig. 73: New gradient coil cover

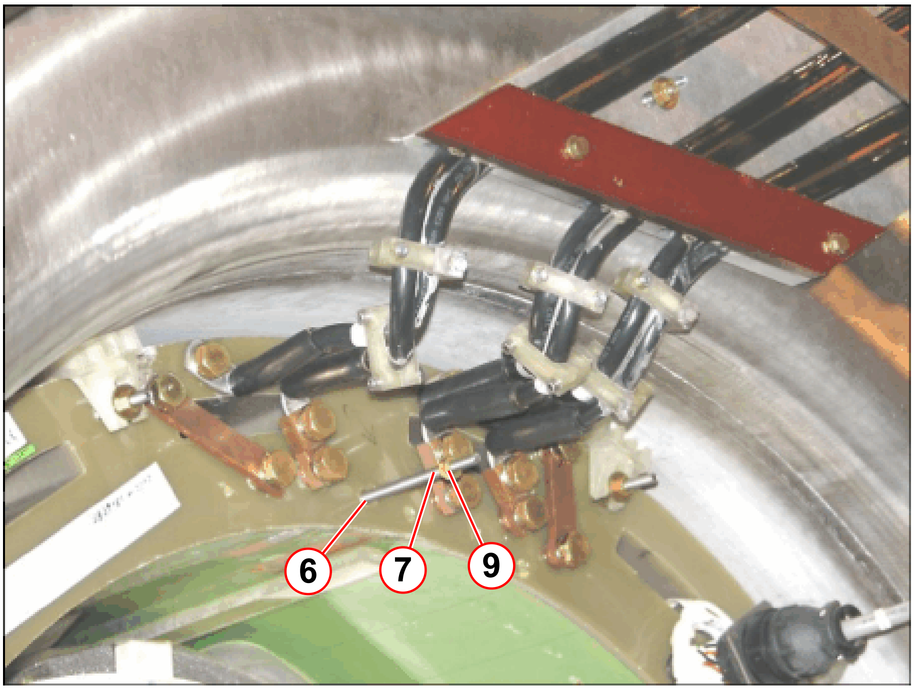


(1) Protective cover
(2) Warning label
(3) Threaded rod, M6*100
(4) Hex nut M6
(5) Disk washer, 6.4
(6) Disk washer, 6.2*18*2

1. Replace the 60 mm long threaded rod with the new 100 mm long threaded rod (→ 3/Fig. 73 Page 103) from the spare parts kit. Apply a drop of Loctite to the thread to secure it.
2. Screw the threaded rod to the gradient coil until it stops (→ 6/Fig. 74 Page 104).

3. Attach a M6 brass nut (using a little bit of Loctite 221) and a new smaller disk washer (item 9 in the kit) to the threaded rod (➔ Fig. 74 Page 104) and (➔ Fig. 75 Page 104).

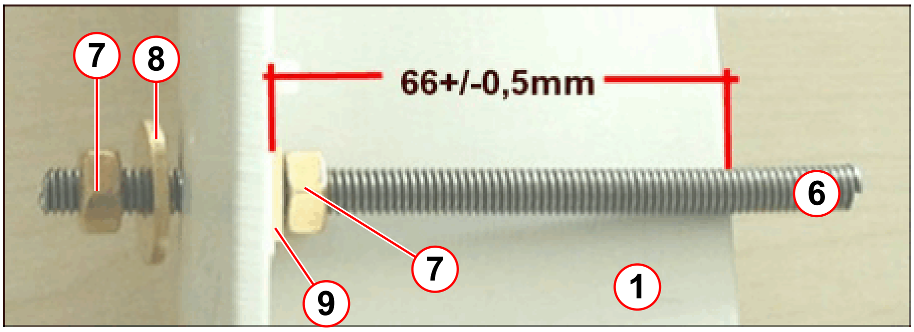
Fig. 74: Threaded rod



(6) 100 mm threadedrod
(7) Hex nut M6
(9) Disk washer 6,2 mm

4. Turn the M6 hex nut (➔ 7/Fig. 75 Page 104) to achieve a distance of 66 +/- 0.5 mm.

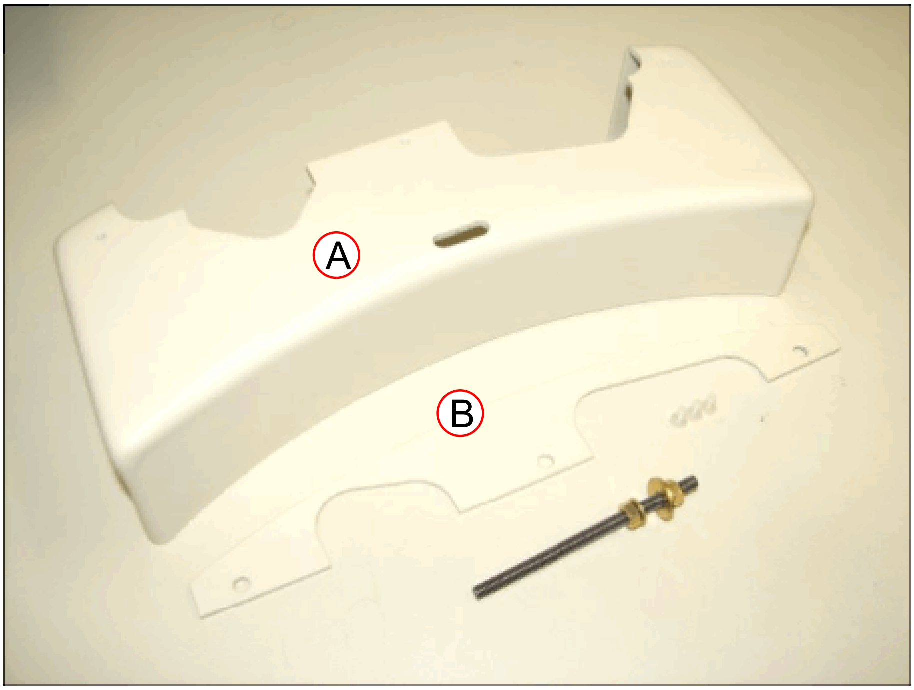
Fig. 75: New threaded rod with protection cover



(1) Protection cover
(6) M6*100 mm threaded rod
(7) M6 hex nut
(8) Disk washer 6,2*18
(9) Disk washer 6,2

5. If not performed previously, disassemble (3 slotted nylon cylinder head screws) the new protective cover (➔ Fig. 76 Page 105). Extra nylon screws have been provided in the kit.

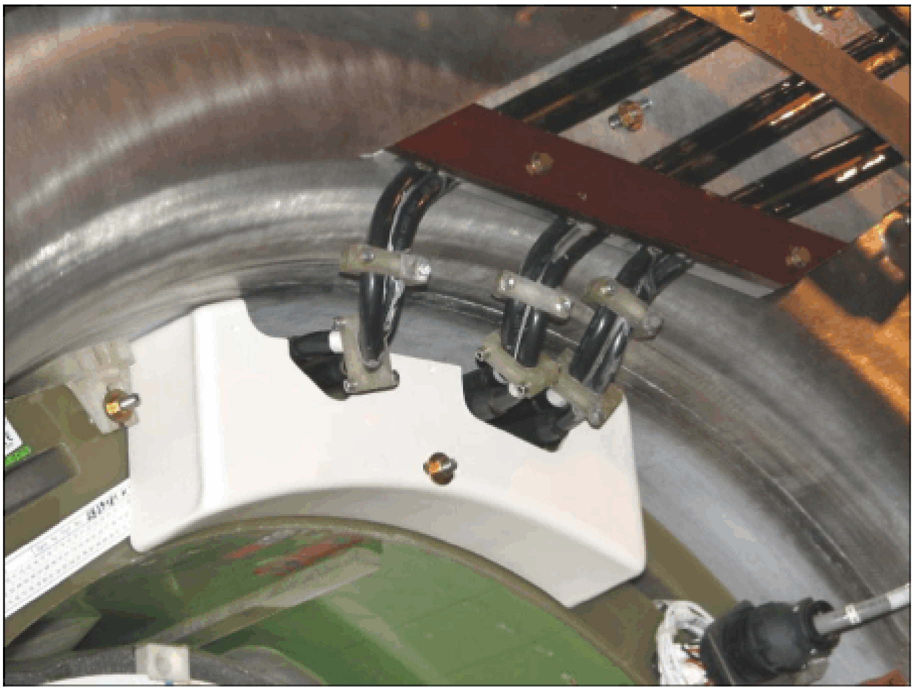
Fig. 76: Lid



(1) Part A
(2) Part B

6. Fasten the new protective cover (→ A/Fig. 76 Page 105) with the M6 hex nut (→ 7/Fig. 75 Page 104) and washer (→ 8/Fig. 75 Page 104) to the 100 mm threaded rod (→ Fig. 77 Page 106).

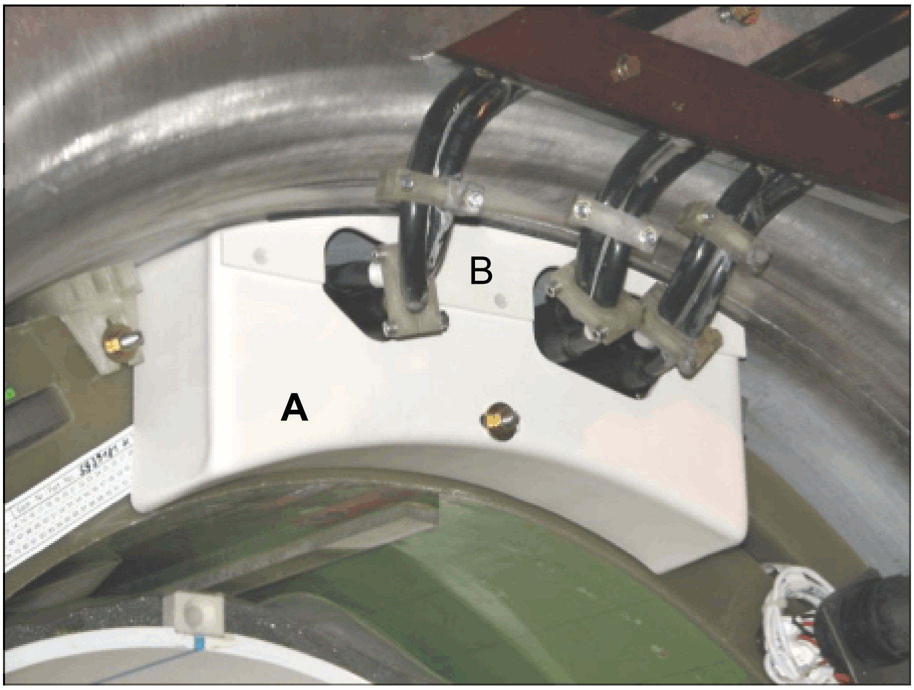
Fig. 77: Lid part A fastened



(1) Part A

7. Fasten part B (→ Fig. 78 Page 106).

Fig. 78: Lid completed

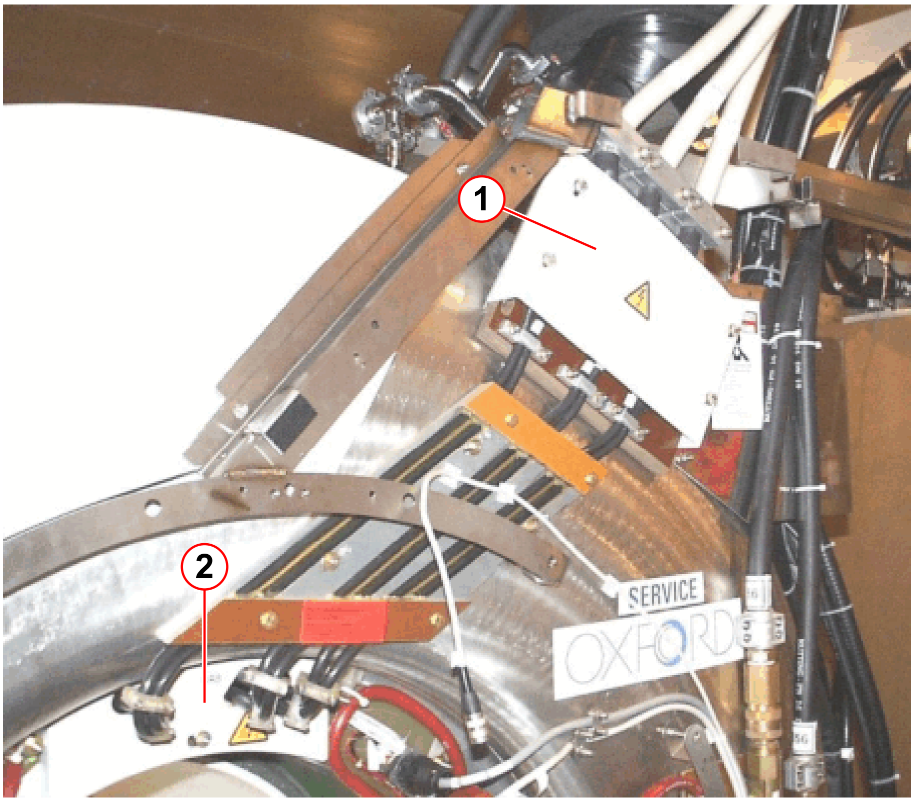


(1) Part A
(2) Part B

8. If you have not already done so, affix the warning label (➔ 2/Fig. 73 Page 103) to the new lid.

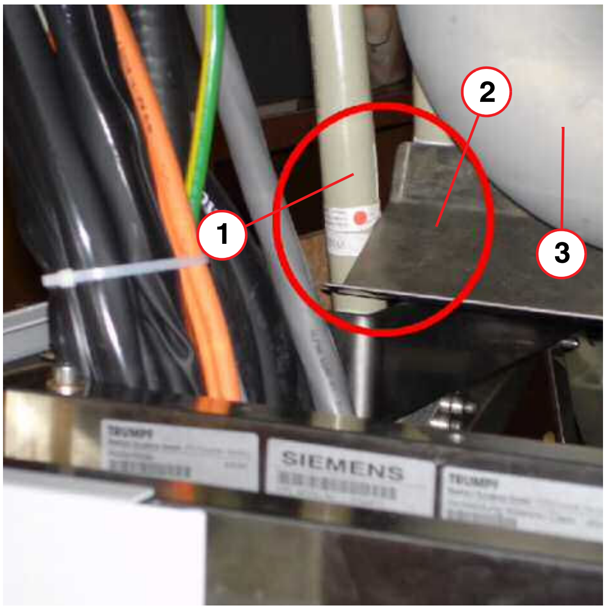
- Fit the lid for the external gradient cables (➔ 1/Fig. 79 Page 107).

Fig. 79: Two covers installed



(1) Connection plate cover
(2) Gradient coil cover (previous style shown in this image)

Fig. 80: Cable routing



(1) Gradient cable
(2) Trip tray
(3) Quench pipe

- Check the routing of the gradient cable.
- The gradient cable should not touch the drip tray.

10.1.4 Final Work Steps/ QA Tests

- Reinstall the magnet covers in reverse order.
- Switch on the circuit breakers for GPA in the ACC cabinet.

Perform Quality Assurance:

- QA/Expert mode/Spike Check
- Run the Spike Check five times
- Image Orientation Check

Version 04

- Current Sensor Unit updated (➔ Current Sensor Assembly / Page 36).
- Add VLM device in the FRU list (➔ FRU Overview / Page 6).
- All: Editorial changes.

Version 05

- Chapter (LF Filters and Blowers) - replaced by chapter (➔ Chokes & Blowers / Page 77). Hint added: Check the function of the blowers after a choke was exchanged.
- All chapters - error correction.
- All chapters - minor editorial changes.

12 List of Hazard IDs

There are no Hazard IDs in this document.

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